

§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

- 本章要求/Requirement of this section
- 电磁兼容性基本概念/ Basic concept of Electro–Magnetic Compatibility (EMC)
- EMC 电控单元干扰源、耦合通道和敏感设备分析/Analysis of the interferential source, coupling channel and sensitive device on ECU
- 电控单元电磁兼容性设计/ The EMC design on ECU?
- 什么是电磁兼容？ /What is EMC?

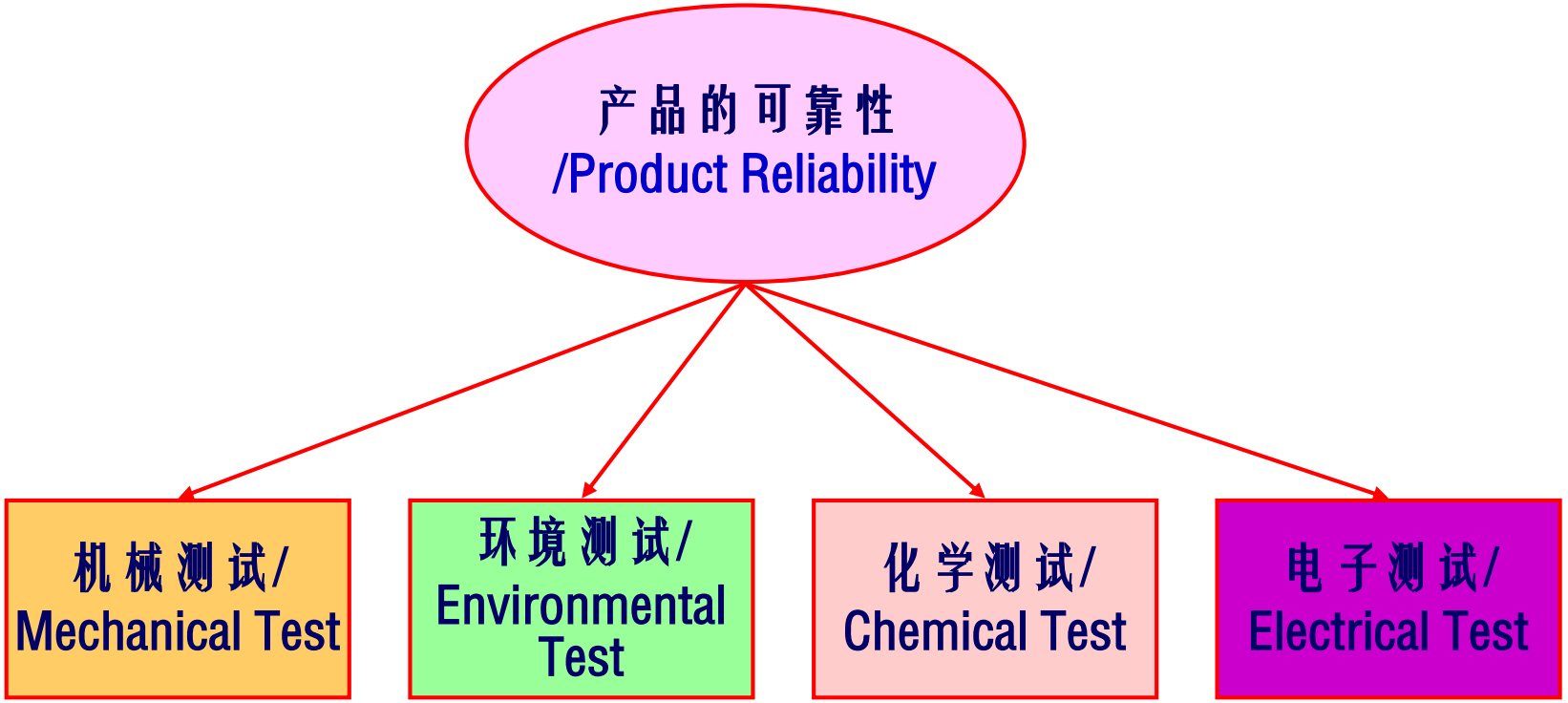
电磁兼容是一种能力，它在其电磁环境中能完成它的功能，而不至于在其环境中产生不能容忍的干扰——国际电工委

/ EMC is the ability of an equipment or system to function satisfactorily in its electromagnetic environment without introducing intolerable electromagnetic disturbances to anything in that environment –International Electric Committee (IEC).



§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

- 为什么要考虑EMC? /Why consider EMC problem?
  - ◆产品可靠性/ Product Reliability





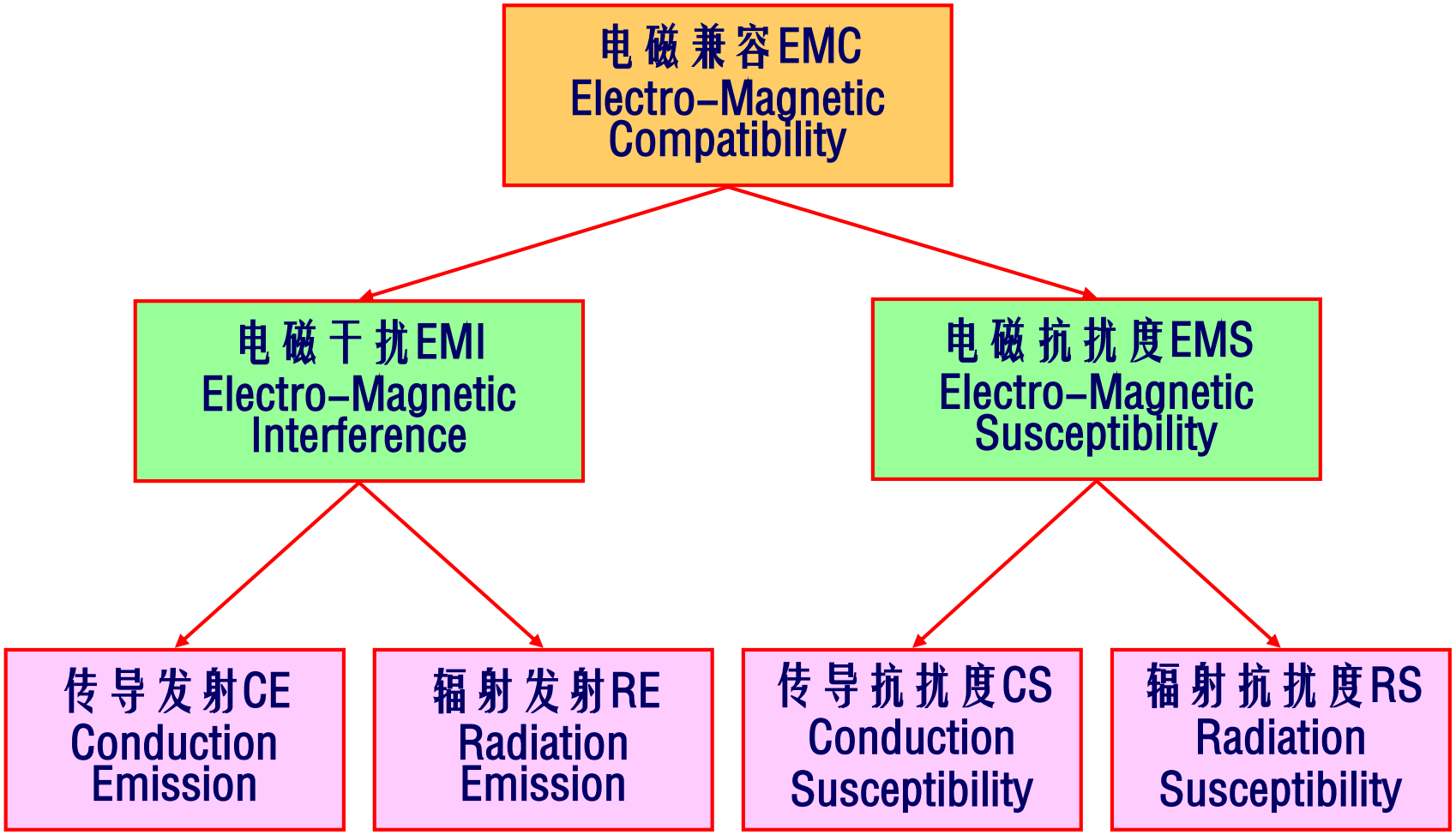
§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

- EMC 的三个要素/ Three Elements
- 解决电磁兼容的问题/Solve EMC Problem: Improve three elements.

干扰源 Interferential Source	→ 传播途径 Couple Channel	→ 敏感设备 Sensitive Device
传染病源 Interferential Disease	→ 传播途径 Couple Approach	→ 易感人群 Sensitive People



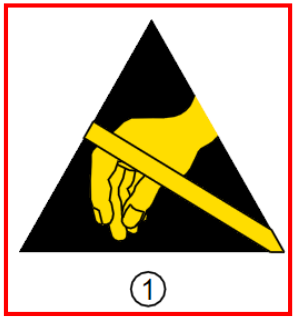
●EMC 的专业名词/ EMC Conceptions

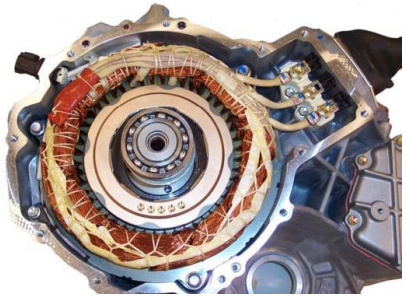




§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

- 车内干扰源/ Interference inside vehicle
  - ◆带有控制开关的电感性电气设备干扰（辐射和传导）
    Inductive electric device with power switch (Radiation and Conduction)
  - ◆无线电发射设备干扰（辐射）
    Interference generated by wireless emission device (Radiation)
  - ◆脉冲数字电路和开关电路干扰（辐射和传导）
    Interference generated by pulse digital circuit and switching circuit (Radiation and Conduction)
  - ◆旋转设备干扰（辐射）
    Interference generated by rotational device (Radiation)
  - ◆静电放电干扰（辐射和传导）
    Electro–Static Discharge (ESD) (Radiation and Conduction)











车外干扰源/ Interference outside vehicle

1.无线电设施的射频干扰（辐射）/ Wireless radio frequency interference (Radiation)

2.交通设施干扰（辐射）/ Interference generated by transportation device (radiation)

3.其它自然干扰源（辐射）/ Other natural interferential source

◆例如雷电–太阳黑子–太空粒子，……

❖总之，du/dt或 di/dt大的地方就是干扰源。如：雷电、继电器、可控硅、电机、高频时钟等/ In summary, the interference is related with du/dt, di/dt, or dΦ/dt such as thunder, relay, SCR, motor or high frequency clock source and so on.











§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

ECU 内部的干扰源/ Interference source inside ECU

1. 电感性负载的驱动电路噪声（辐射和传导）/ Noise generated by Inductive Load drive circuit (Radiation and conduction)
2. 高频信号线上噪声（传导）/ Noise generated by the high frequency signals
3. 电控单元的接插件引入噪声（传导）/ Noise induced by the input plug (Conduction)
4. 电源系统的噪声（传导）/ Noise generated by power supply (Conduction)
5. 电控单元内部的高频电路噪声（辐射）/ Noise generated by the high frequency circuit inside ECU



§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

- 电控单元敏感设备/ The Sensitive Device in side ECU
  - ◆模拟信号通道/ The analogue signals
  - ◆数字集成电路/ The digital IC circuit
  - ◆单片机/ The MCU

- 解决EMC的方法/ The methods of deal with EMC problem
  - ◆抑制干扰源；/ Control the interference
  - ◆切断干扰传播路径；/ Cut off the transfer approach (Radiation and Conduction)
  - ◆提高敏感器件的抗干扰性能/Improve Immunity of the sensitive device



§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

（一）抑制干扰源就是减小干扰源的du/dt，di/dt和dΦ/dt / Prevent or smooth the interference's du/dt, di/dt and dΦ/dt

（1）感性负载的线圈增加续流二极管，消除断开线圈时产生的反电动势干扰。/Add the flywheel diode for inductive loads to eliminate the inductive voltage when coil is switch off.

（2）在大的感性负载两端并接火花抑制电路（一般是RC串联电路，电阻一般选几K到几十K，电容选0.01uF），减小电火花影响；/ Parallel a spark prohibit circuit by using RC to minimize the influence of switch off

（3）给电机线圈加滤波电路，注意电容、电感引线要尽量短/ Add a filter circuit for the winding of motor. The connection between the capacitor and conductance should be sufficient short.

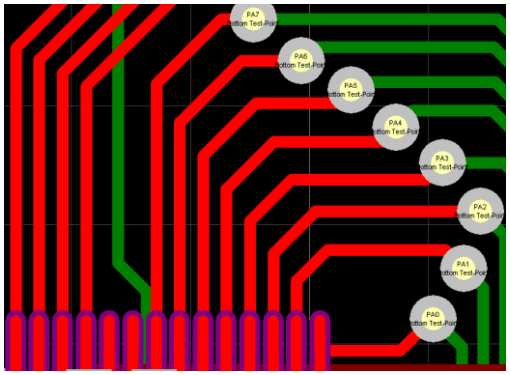


（4）电路板上每个IC要并接一个0.01μF~0.1μF高频电容，以减小IC对电源的 影响。注意高频电容的布线，连线应靠近电源端并尽量粗短，否则，等于增大了电容的等效串联电阻，会影响滤波效果；

Every IC should parallel one 0.01uF to 0.1uF high frequency capacitor in supply power to eliminate the influence of the source. The high frequency as close to power pin as possible to reduce the capacitive effect.

（5）布线时避免90度折线，减少高频噪声发射；

Do not use 90 degree track when design PCB, because that will increase high frequency emission.



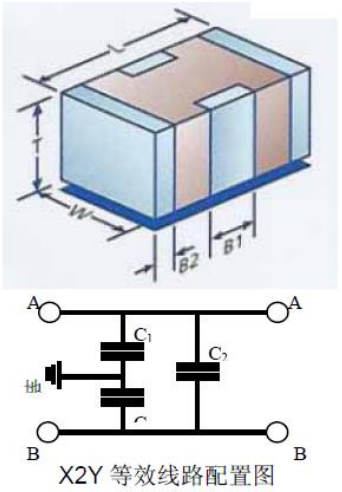


（二）切断干扰传播路径的常用措施如下：

（1）充分考虑电源对ECU的影响。比如，可以利用磁珠和电容组成π形滤波电路，甚至干脆采用DC/DC的隔离电源；

The power supply is the main interference for ECU. It is necessary to use a π type filter to remove the noise, sometimes even use the isolated DC/DC power supply to eliminated interference.

源阻抗	滤波器连接关系	负载阻抗
低		低
低		高
高		低
高		高

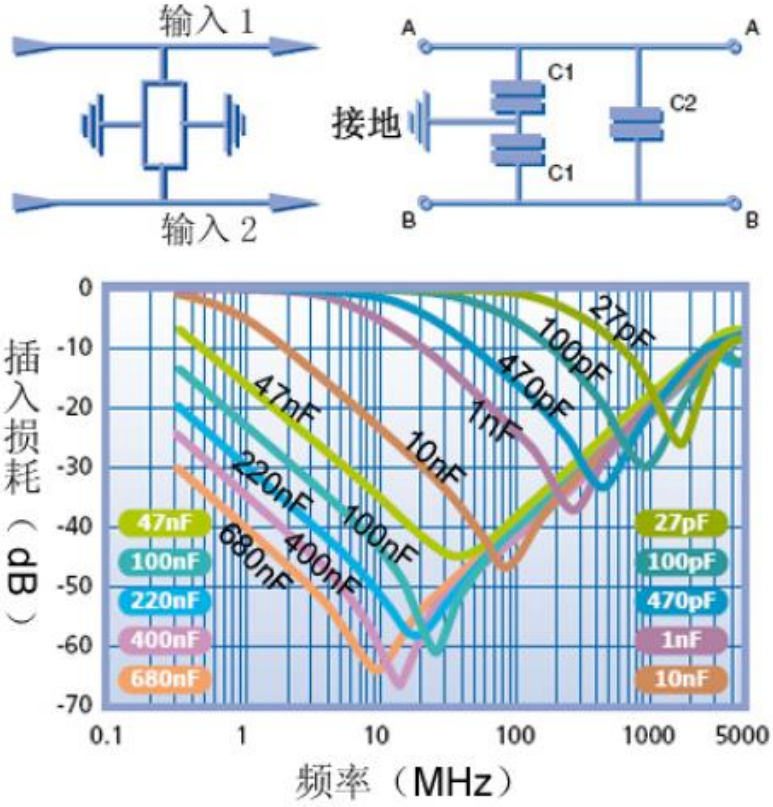


X2Y 等效线路配置图

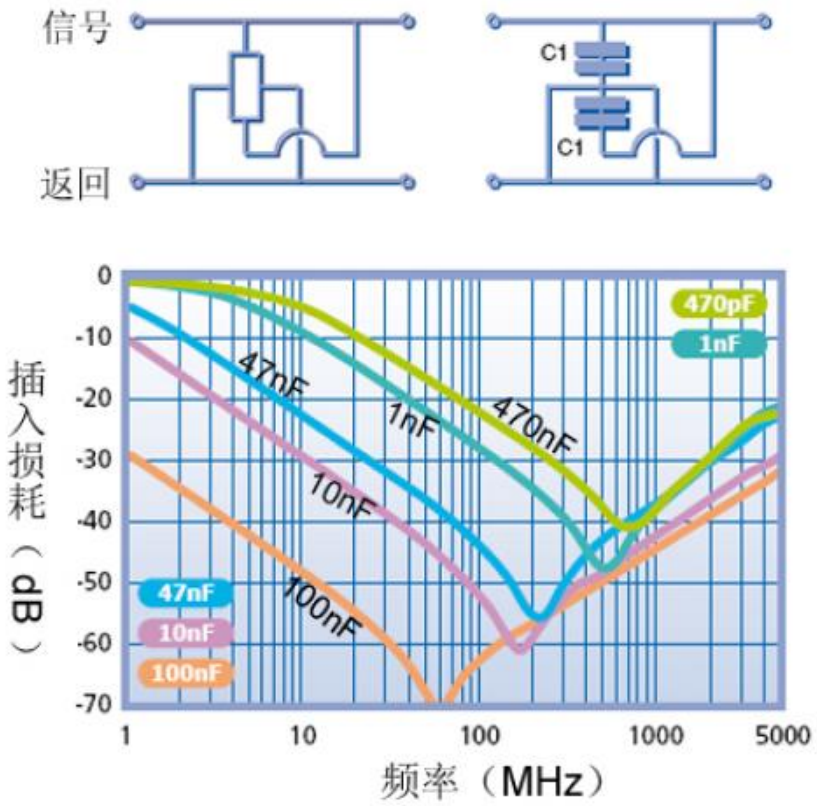


（二）切断干扰传播路径的常用措施如下：

滤波应用



去耦应用



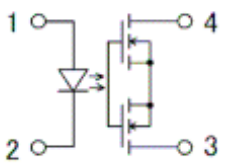


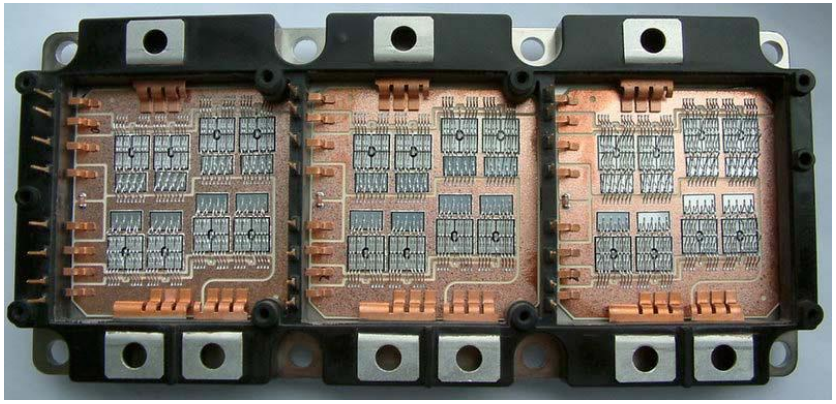
§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

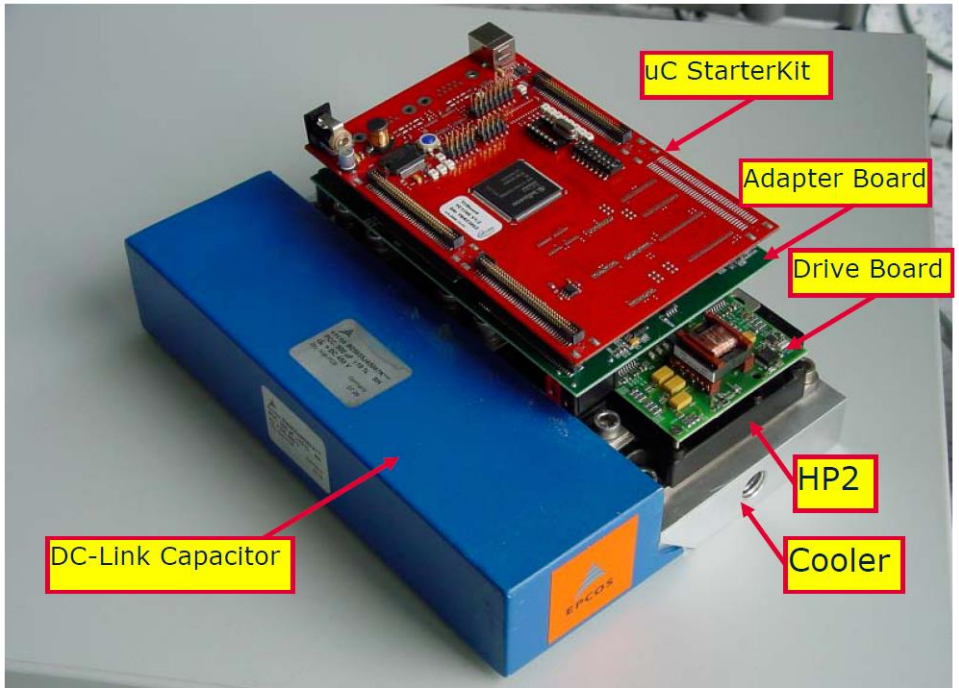
（二）切断干扰传播路径的常用措施如下：

（2）如果单片机的I/O口用来控制电机等噪声器件，在I/O口与噪声源之间应加隔离（增加π形滤波电路）；Sometimes the I/O signal should be isolated with the drive circuit of motor by using π type filter.









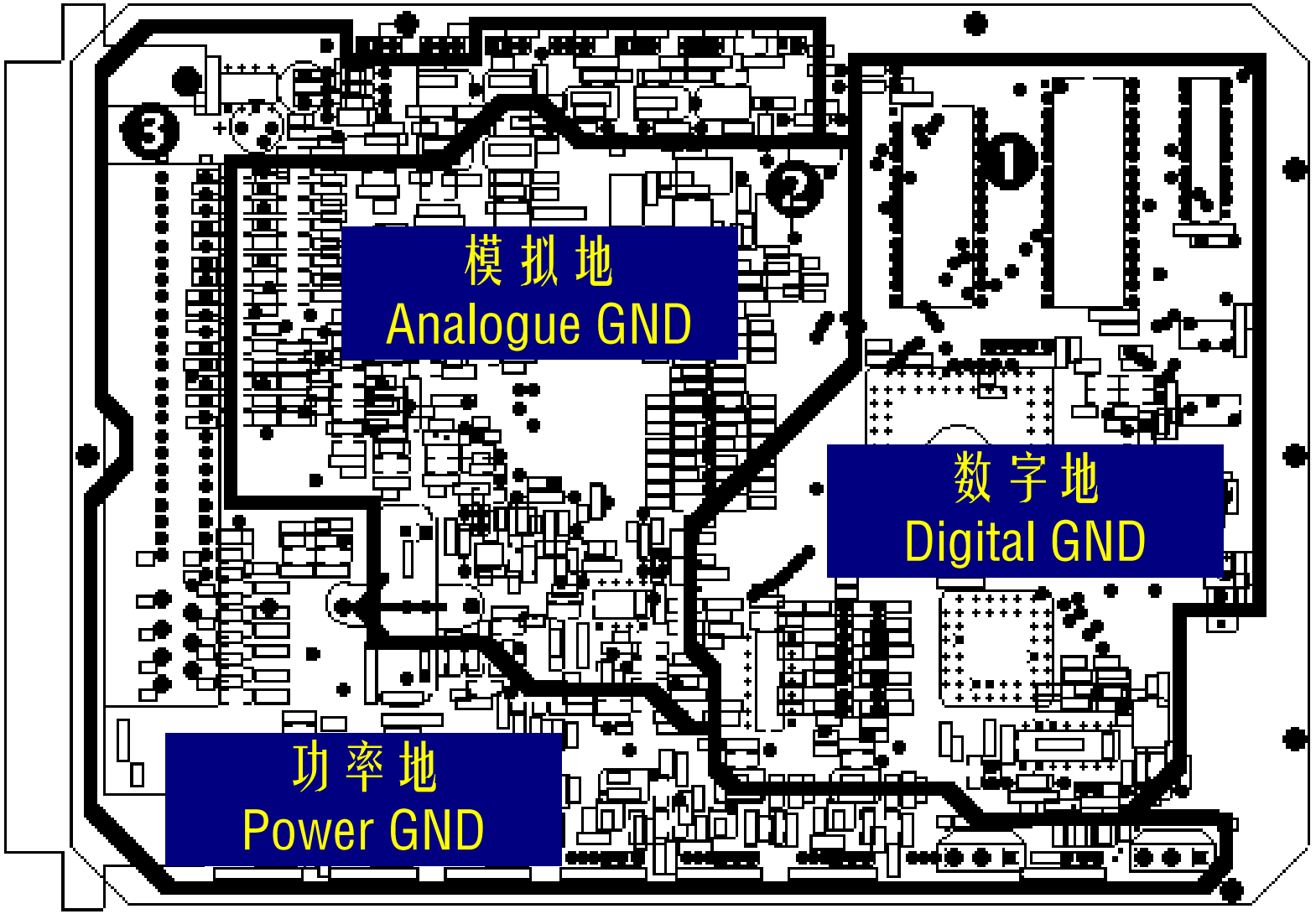


（3）注意晶振布线。晶振与单片机引脚尽量靠近，用地线把时钟区隔离起来，晶振外壳接地并固定；/ The pins of the crystal should as close as possible to the pin of MCU, mean while the clock circuit should be surrounded by GND. The shell of the crystal should connected to ground to reduce of emission.

（4）电路板合理分区，如强、弱信号，数字、模拟信号；尽可能把干扰源（如电机，继电器）与敏感元件（如单片机）远离；

The PCB should be careful to divided into reasonable parts, such as strong signal areas or weak signal areas, digital signal areas or analogue signal areas. Make the interference is as possible as far away from the sensitive circuit.

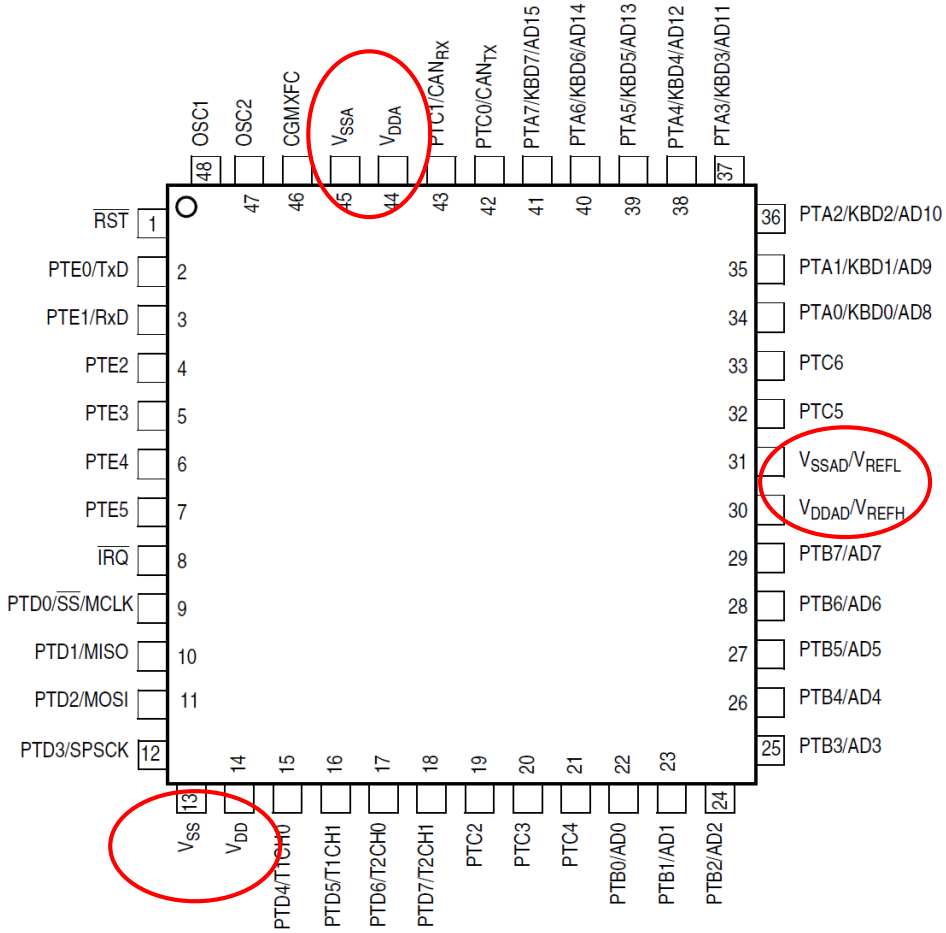



模拟地
Analogue GND
数字地
Digital GND
功率地
Power GND



§4.3 电磁兼容性分析与设计 / EMC Analysis and Design

（5）用地线把数字区与模拟区隔离，数字地与模拟地要分离，最后在一点接于电源地；A/D、D/A芯片布线也以此为原则，厂家分配A/D、D/A芯片引脚排列时已考虑此要求；/Separate the digital and analogue area by corresponding GND, the GNDs should be separated and connected to the COMMON potential in one point. For example the A/D or D/A ICs has been considered this point when design.



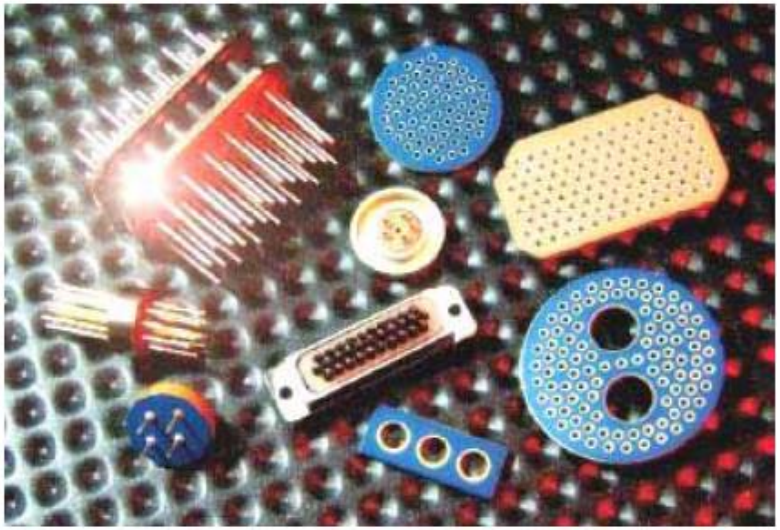


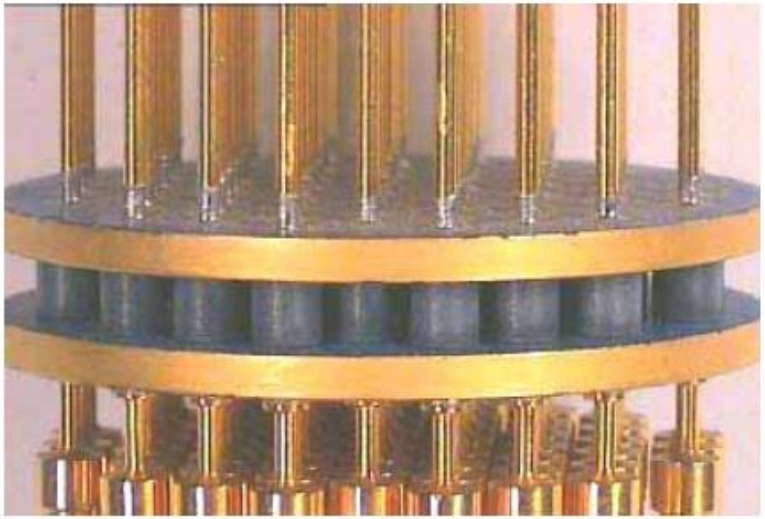
（6）单片机和大功率器件的地线要单独接地，以减小相互干扰。大功率器件尽可能放在电路板边缘； / The ground of MCU should be separate with the ground of power driver circuit in one point. The large power component should be placed in the edge of the PCB.

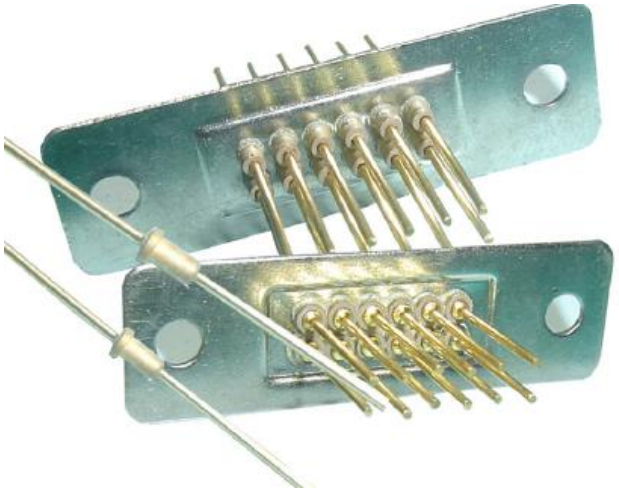
（7）在单片机I/O口，电源线，电路板连接线等关键地方使用抗干扰元件 如磁珠、磁环、电源滤波器，屏蔽罩以及完全隔离器件等。/ It is recommended to use shield or filter component in the power supply or connections to the MCU IO ports.



（二）切断干扰传播路径的常用措施如下：









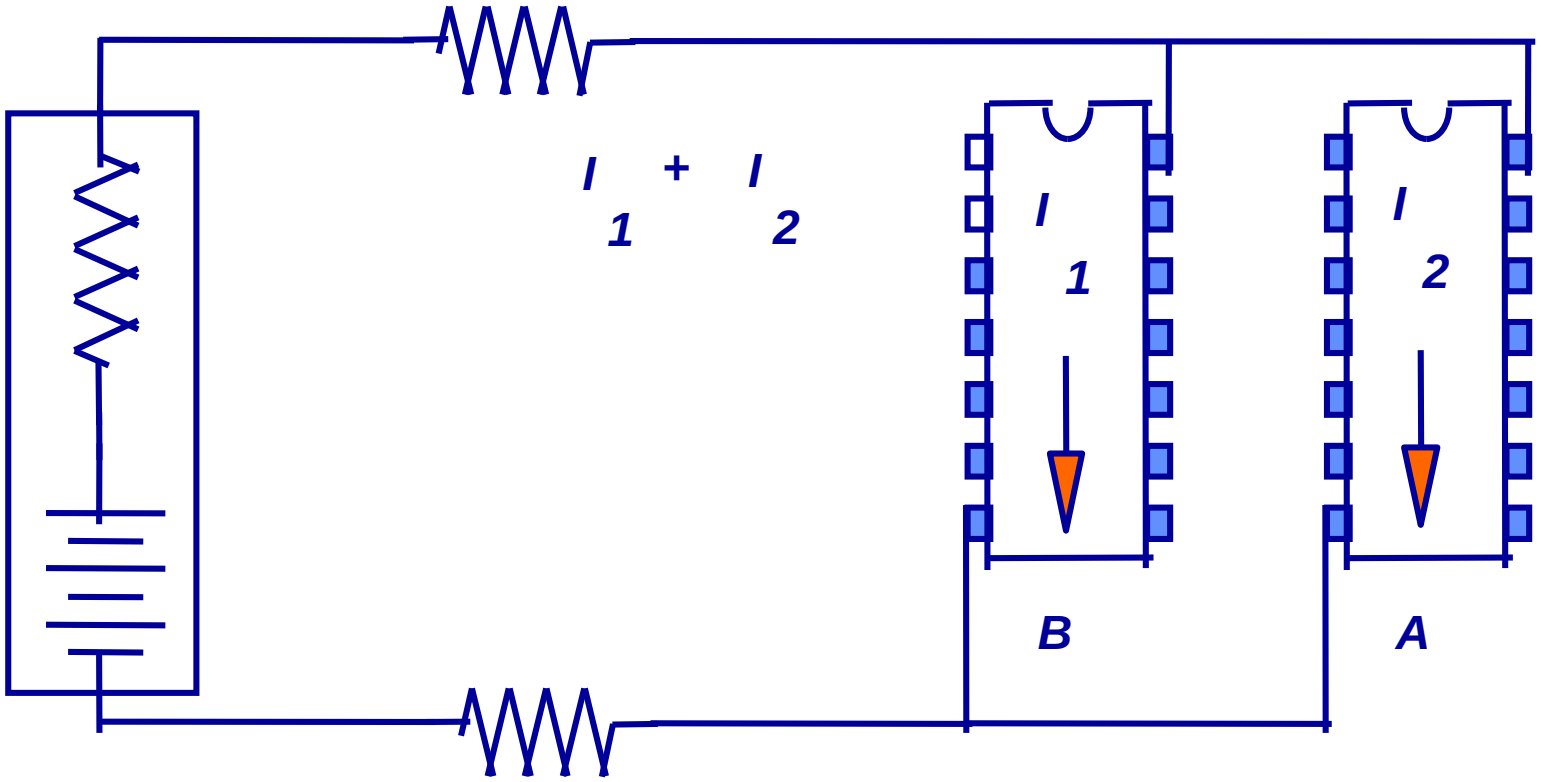


（三）提高敏感器件的抗干扰性能 / Improved the immunity of sensitive device.

（1）布线时尽量减少回路环的面积，以降低感应噪声；/ Try to minimize the circuit loop area when place the track. This is effetive to reduce the indutive noise.

（2）布线时，电源线和地线要尽量粗。除减小压降外，更重要的是降低耦合噪声；/ The tracks of power and ground should be as thick as possible, it is helpful to reduce voltage drop along the track and more important thing is to reduce the coupling noise.

●电源系统回路/ power supply loop for IC





（3）对于单片机闲置的I/O口，不要悬空，要接地或接电源。其它IC的闲置端在不改变系统逻辑的情况下接地或接电源；/ The non-used ports of MCU should not be empty, they should be connected to power supply or ground; The other IC non-used port also needs connected to power or ground without changing the system logic state.

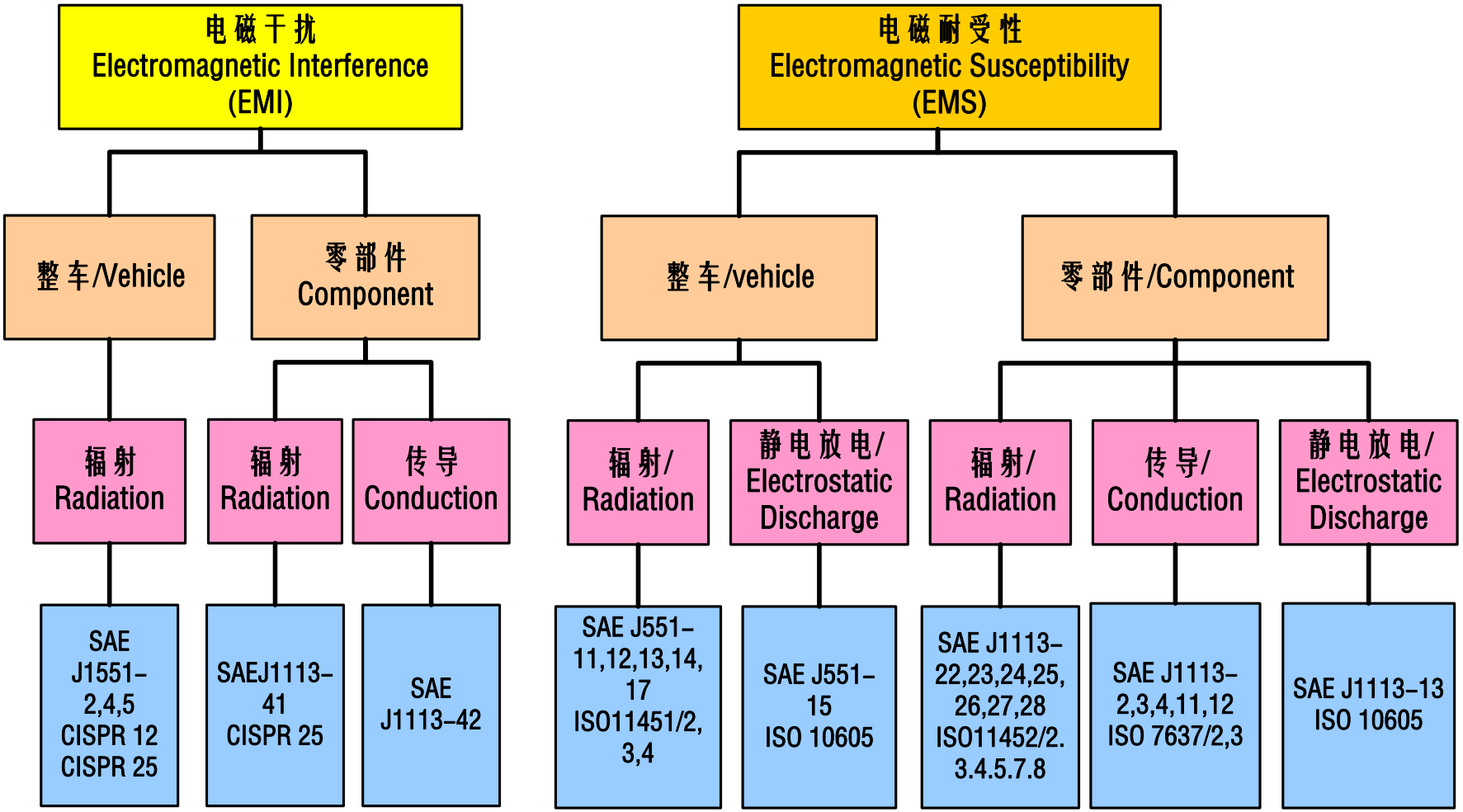
（4）对单片机使用电源监控及看门狗电路，如：IMP809，IMP706，IMP813，X25043，X25045等，可大幅度提高整个电路的抗干扰性能，受干扰之后可以迅速恢复； / Add necessary power monitor of watchdog circuit, for example IMP809 or so on to improve the immunity performance of the whole system significantly or ensure the system could recovery quickly after interference.

（5）在速度能满足要求的前提下，尽量降低单片机的晶振和选用低速数字电路；/Try to decrease the crystal frequency when the control speed can fulfill the requirement.

（6）IC器件尽量直接焊在电路板上，少用IC座。/ Try to eliminate the IC plug and weld the IC directly on the PCB.



●EMC 的测试规范/ EMC Standards





●各种电磁兼容标准比较 / EMC Standards Comparison

标准号 Standards	试验对象 Test object		干扰/抗扰 Interference /Susceptibility		传播方式 /couple mode		
	整车 Vehicle	零部件 Component	干扰 Interference	耐受性 Susceptibility	传导 Conduction	辐射 Radiation	静电放电 Electro Static Discharge
ISO11451	√			√		√	
ISO11452		√		√		√	
ISO7637		√		√	√		
ISO10605	√			√			√
CISPR12	√		√			√	
CISPR25	√		√		√	√	
SAE J1113	√		√		√		
SAE J551		√	√		√		



●SAE 电磁兼容标准举例/Introduction of SAE EMC Standard

标准号	标准名称	发布时间
SAE J551-1	Performance Levels and Methods of Measurement of Electromagnetic Compatibility of Vehicles, Boats (Up to 15M), and Machines(50Hz to 18GHz)	2002
SAE J1551-2	Limits and Methods of Measurement of Radio Disturbance Characteristics of vehicles, Motorboats and Spark-ignited Engine-driven Devices	2004
SAE J1551-4	Test Limits and Methods of Measurement of Radio Disturbance Characteristics of vehicles and Devices, Broadband and Narrowband, 150kHz to 1000MHz	2000
SAE J1551-5	Performance Levels and Methods of Measurement of Magnetic and Electric Field Strength from Electric Vehicles, Broadband, 9kHz to 30Mhz	2004



●SAE 电磁兼容标准举例/Introduction of SAE EMC Standard

标准号	标准名称	发布时间
SAE J1551-11	Vehicle Electromagnetic Immunity — Off-vehicle Source	2000
SAE J1551-12	Vehicle Electromagnetic Immunity — On-board Transmitter Simulation	1996
SAE J1551-13	Vehicle Electromagnetic Immunity — Bulk Current Injection (BCI)	2003
SAE J1551-14	Vehicle Electromagnetic Immunity — Reverberation Chamber	Draft Only
SAE J1551-15	Vehicle Electromagnetic Immunity — Electrostatic Discharge (ESD)	2002
SAE J1551-17	Vehicle Electromagnetic Immunity — Power Line Magnetic Fields	2003



●SAE 电磁兼容标准举例/Introduction of SAE EMC Standard

标准号	标准名称	发布时间
SAE J1113–1	Electromagnetic Compatibility Measurement Procedures and Limits for Components of Vehicles, Boats(Up to 15M), and Machines(Except Aircraft)(50Hz to 18GHz)	2002
SAE J1113–2	Conducted Immunity, 30Hz to 250kHz, Power Leads	2004
SAE J1113–3	Conducted Immunity, 250kHz to 500MHz, Direct Radio Frequency Power Injection	1995
SAE J1113–4	Conducted Immunity–Bulk Current Injection (BCI) Method	2004
SAE J1113–11	Immunity to Conducted Transients on Power Leads	2000
SAE J1113–12	Electrical Interference by Conduction and coupling – coupling Clamp	2000
SAE J1113–13	Immunity to Electrostatic Discharge	2004



●SAE 电磁兼容标准举例/Introduction of SAE EMC Standard

标准号	标准名称	发布时间
SAE J1113-21	Road Vehicles–electrical Disturbances by Narrowband Radiated Electromagnetic Energy–component Test Method Absorber Lined Chamber	1998
SAE J1113-22	Immunity to Radiated Magnetic Fields from Power Lines	2003
SAE J1113-23	Immunity to Radiated Electromagnetic fields,10KHz to 200MHz, Strip Line Method	2000
SAE J1113-24	Immunity to Radiated Electromagnetic Fields,10KHz to 200MHz, TEM Cell Method	2000
SAE J1113-25	Immunity to Radiated Electromagnetic Field, 10KHz to 200MHz Tri-Plate Line Method	1999
SAE J1113-26	Immunity to AC Power Line Electric Fields	2001
SAE J1113-27	Immunity to Radiated Electromagnetic Field – Reverberation Chamber Method	1995
SAE J1113-28	Immunity to Radiated Electromagnetic Field – Reverberation Method(Mode Tuning)	2004

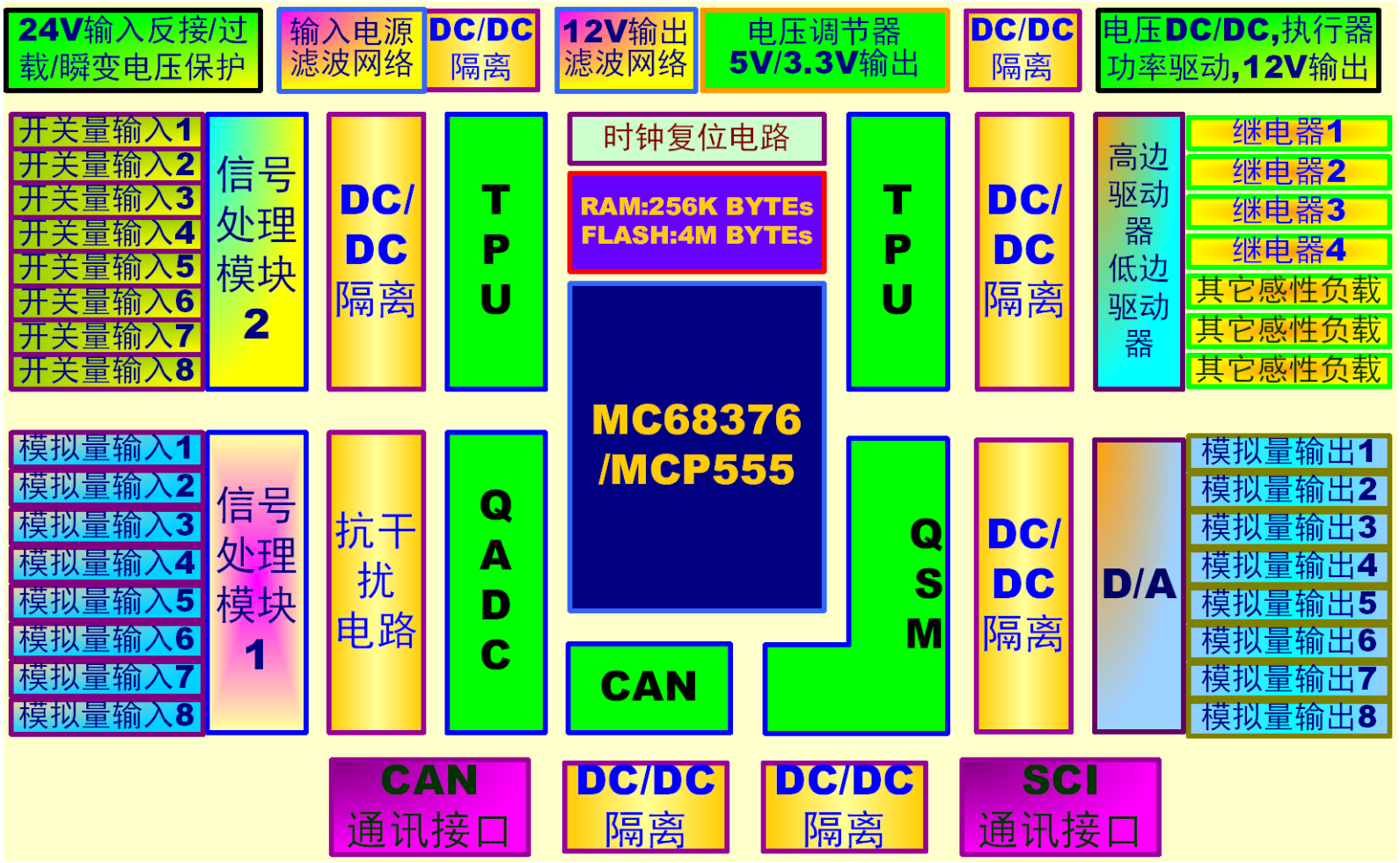


- EMC法规举例/Example of EMC regulation
  - 传递抗扰度：ISO-7637
  - 静电放电抗扰度：ISO-10605



§4.4 ECU设计举例 / Example of ECU Design

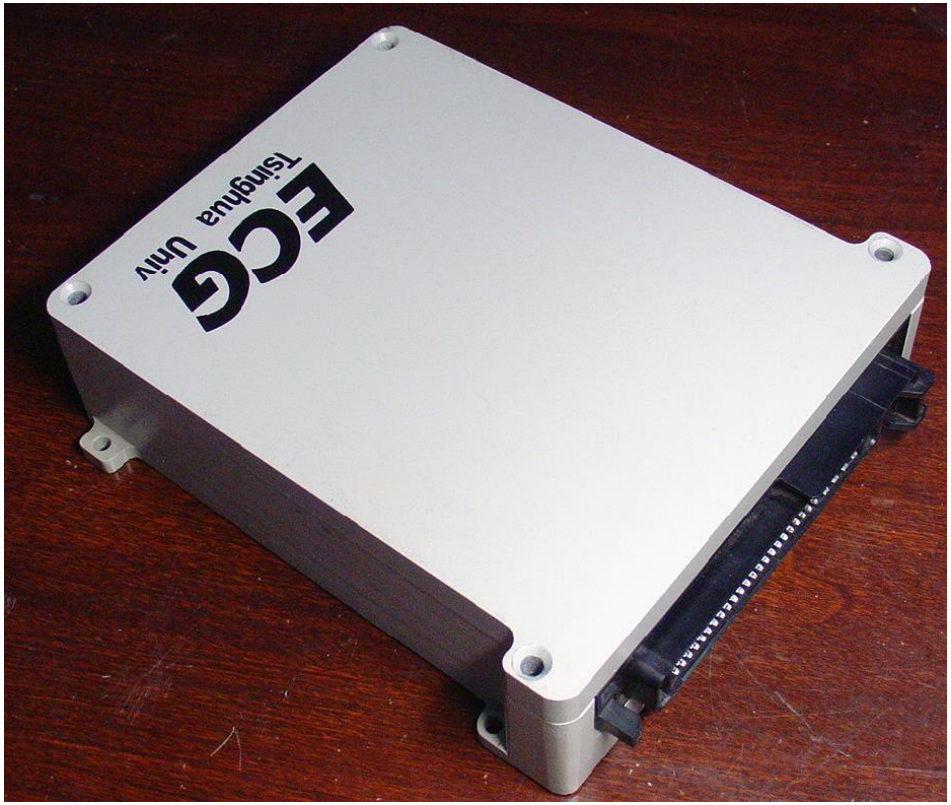
混合动力系统的VCU框架→优先考虑EMC

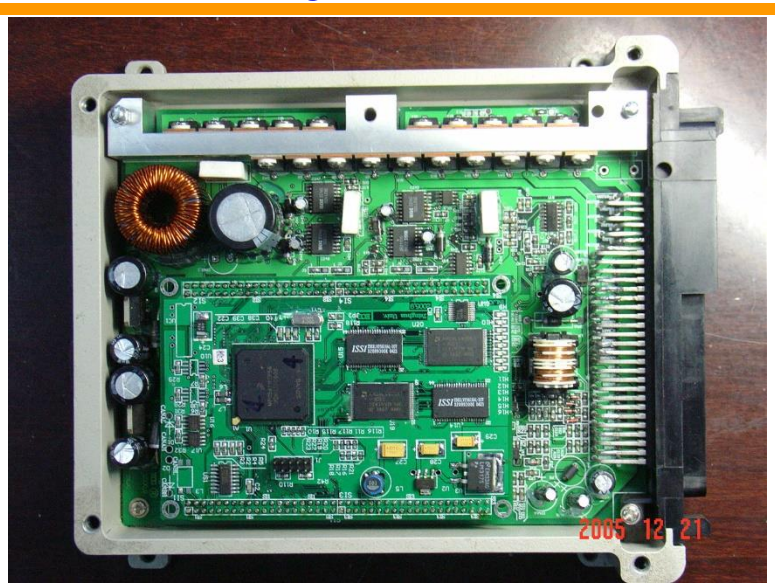




§4.4 ECU设计举例
/ Example of ECU Design

●柴油机ECU
/ Photos of Diesel ECU



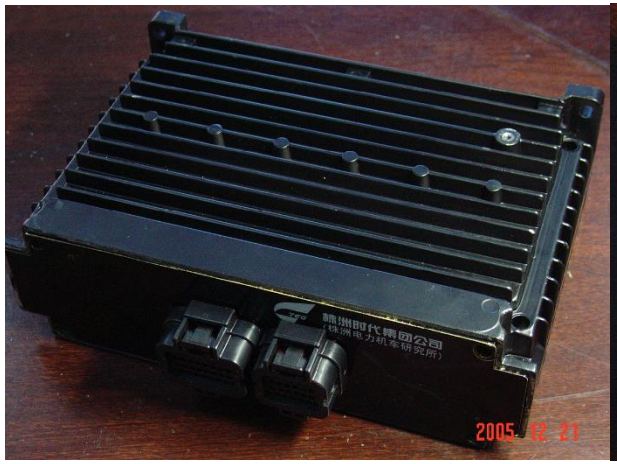


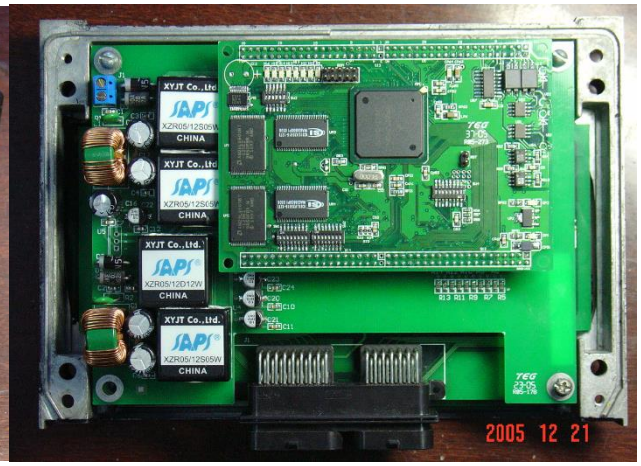


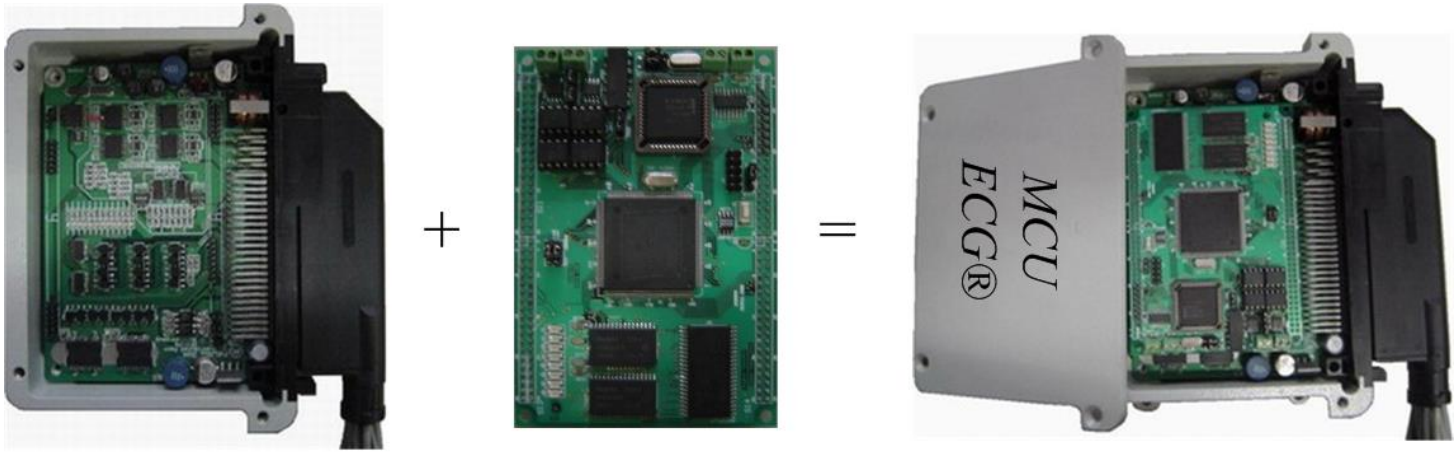


§4.4 ECU设计举例 / Example of ECU Design

- APU ECU and
- FUEL ECU





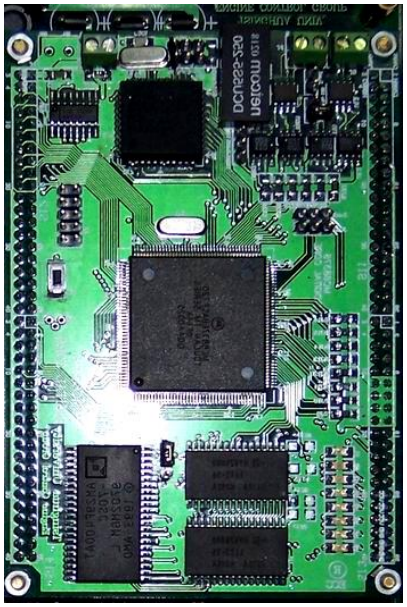


MCU母板 + MCU子板 = MCU/主控制器

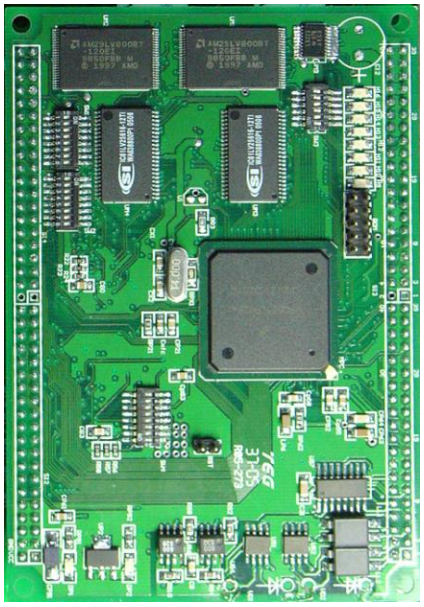


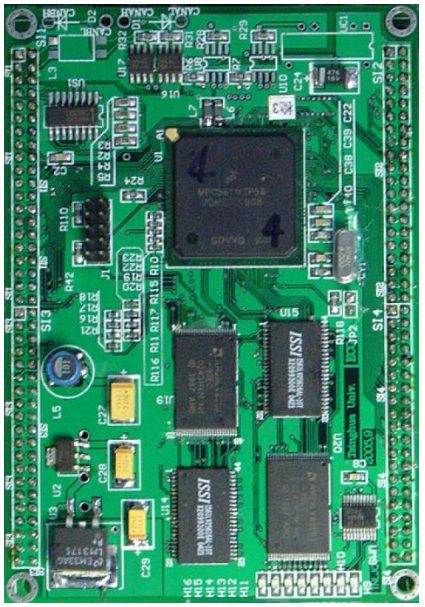
§4.4 ECU设计举例 / Example of ECU Design

- 通用数字核心子板，从左到右/ Digital Core, L2R
- MC68376、MPC555、MPC561、MPC563







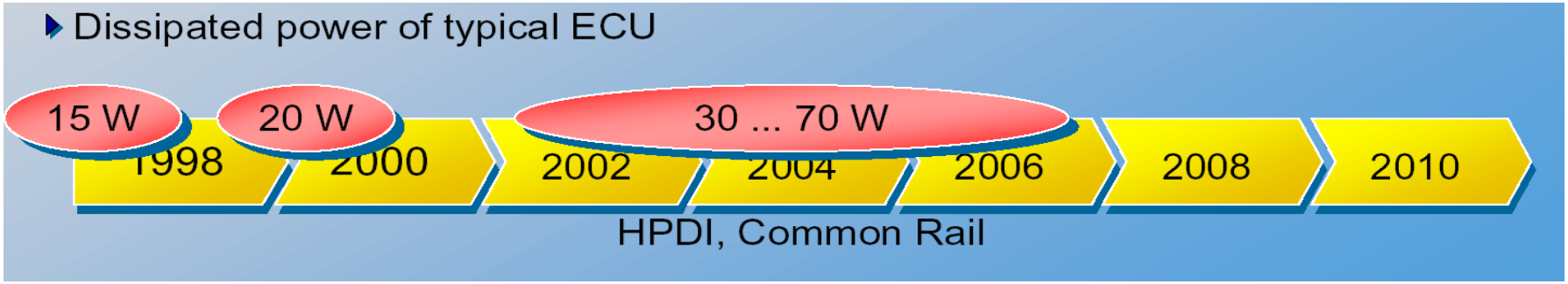




§4.5 ECU的优化设计 /Optimize Design of ECU

❑ 市场竞争/ Market Competition

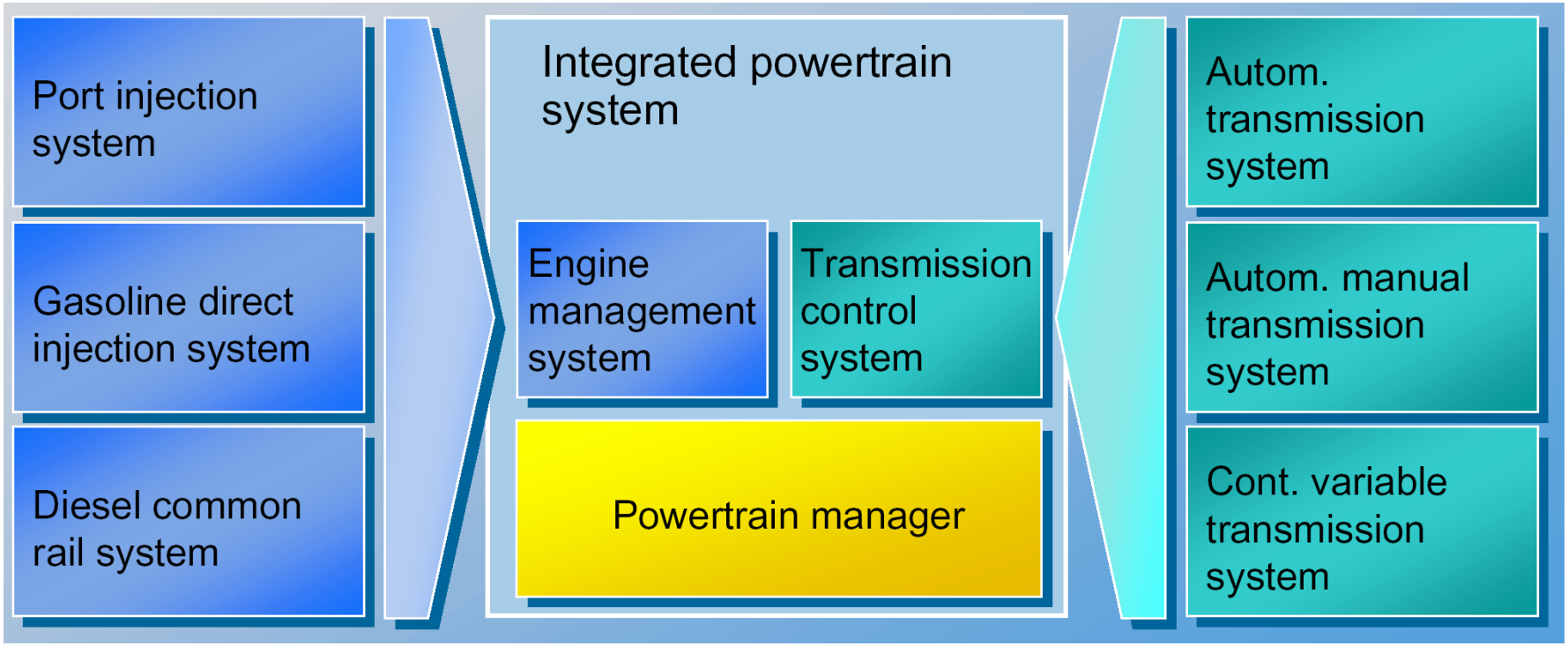
- ➢ 更加严酷的环境/ More critical environment
- ➢ 更加完善的功能/ More complex functions
- ➢ 更高的质量和可靠性/Higher quality and reliability
- ➢ 更低的价格/Lower cost
- ➢ 更小的体积/More compact volume
- ➢ 散热能力更强/Better heat dissipation





§4.5 ECU的优化设计 /Optimize Design of ECU

- 动力系统电子控制/Powertrain Electronic Control
  - →平台化，模块化，通用化/ Platform, Modularize, Generalize

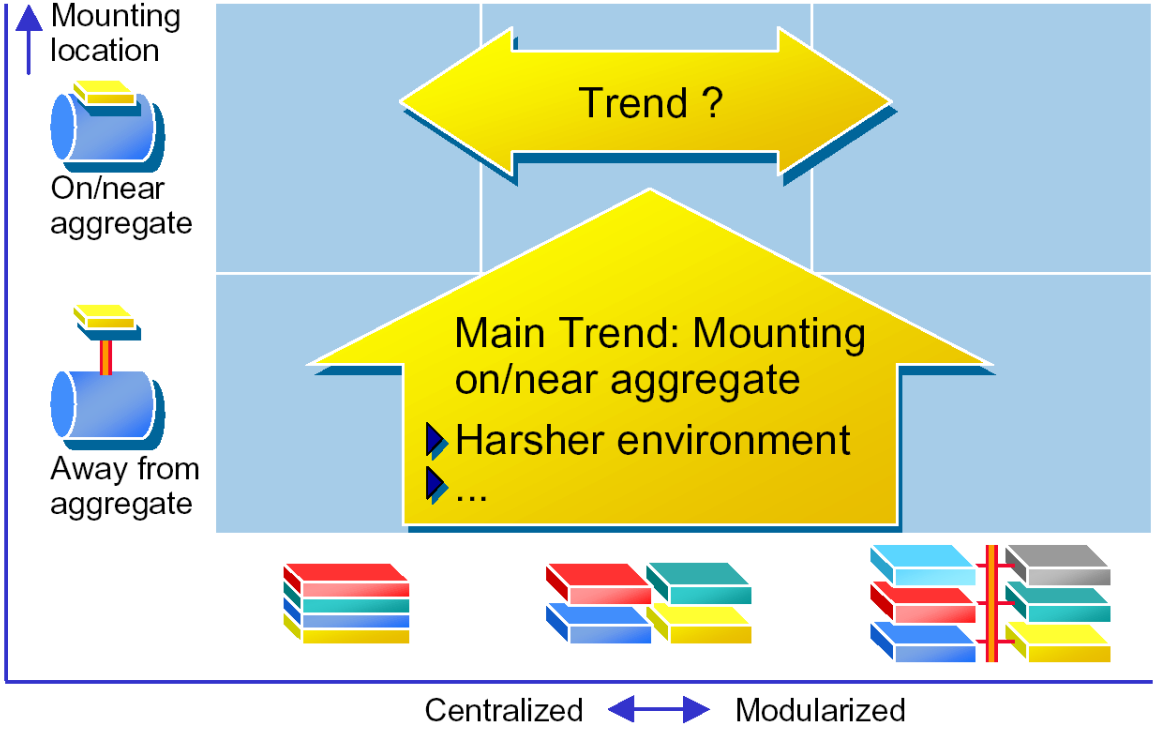


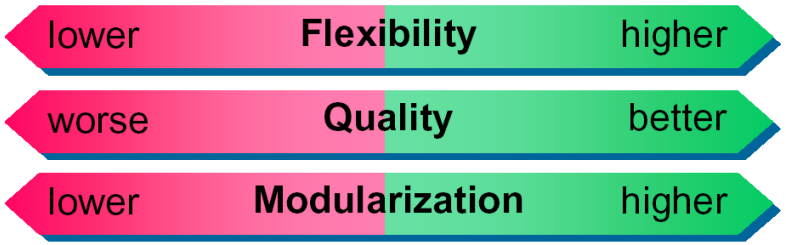


§4.5 ECU的优化设计 /Optimize Design of ECU

❑ ECU的数量和位置布置/ ECU Quantity and Location





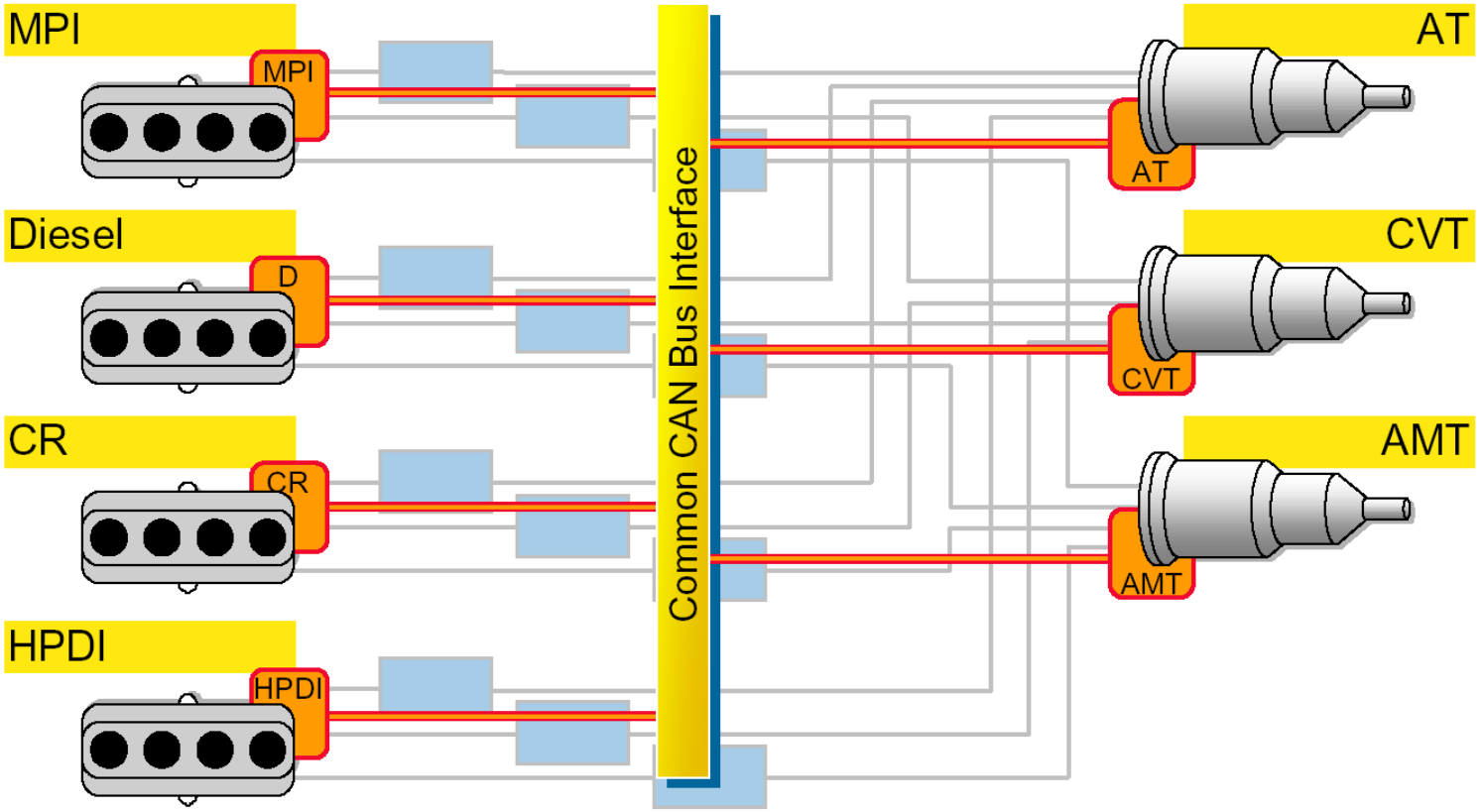




§4.5 ECU的优化设计 /Optimize Design of ECU

❑动力电子控制系统/ Powertrain Electronic Control

→网络化/ Network





§4.5 ECU的优化设计 /Optimize Design of ECU

❑ ECU的可靠性和散热设计/ Reliability and Heat Dissipation Design

Driving Forces

Closer to aggregate:
- Harsh environment
- ...

Increased power density:
- Smaller
- Higher and
- New functionality

Material limits:
- Reliability
- Lifetime

$$T + \Delta T < T_{max}$$

Activities

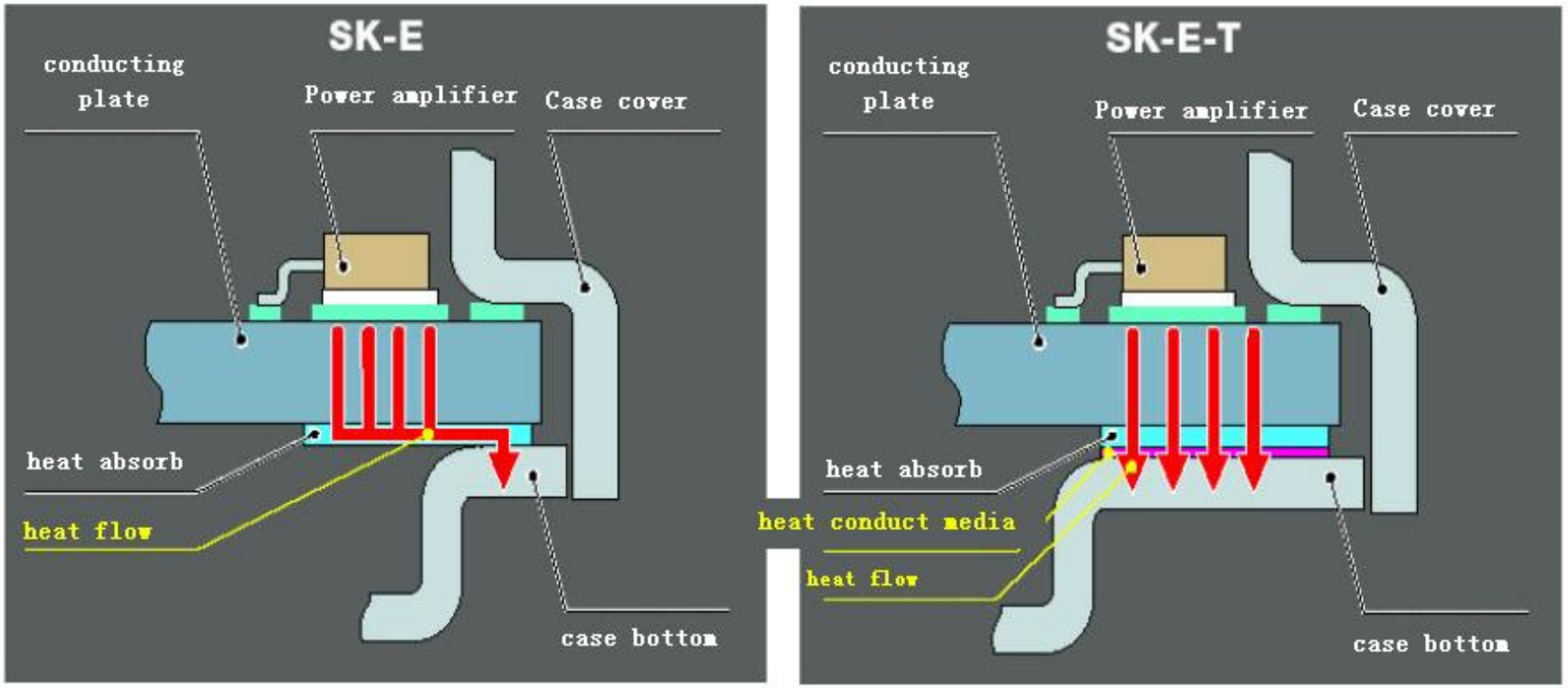
Improved thermal management
- Laminates
- On IAFM
- ...

HT capable materials
- High T_G substrate
- Interconnection
- Components
- ...



§4.5 ECU的优化设计 /Optimize Design of ECU

❑ ECU的可靠性和散热设计/ Reliability and Heat Dissipation Design

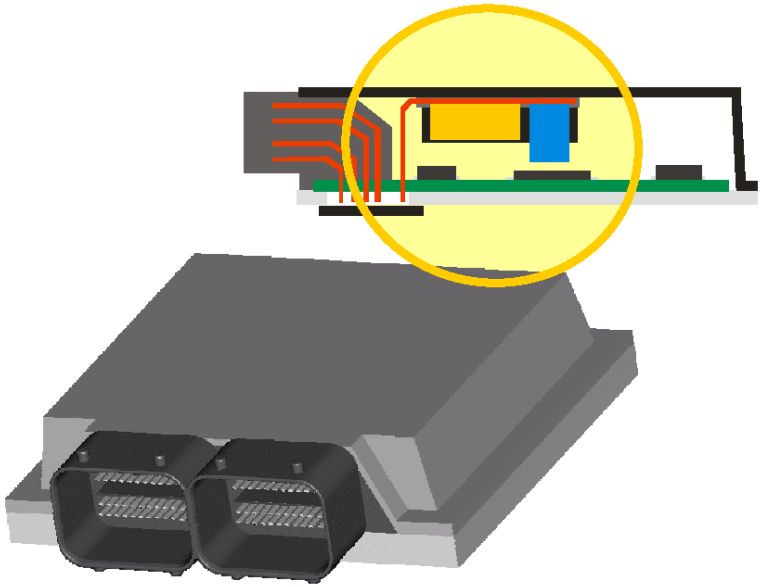


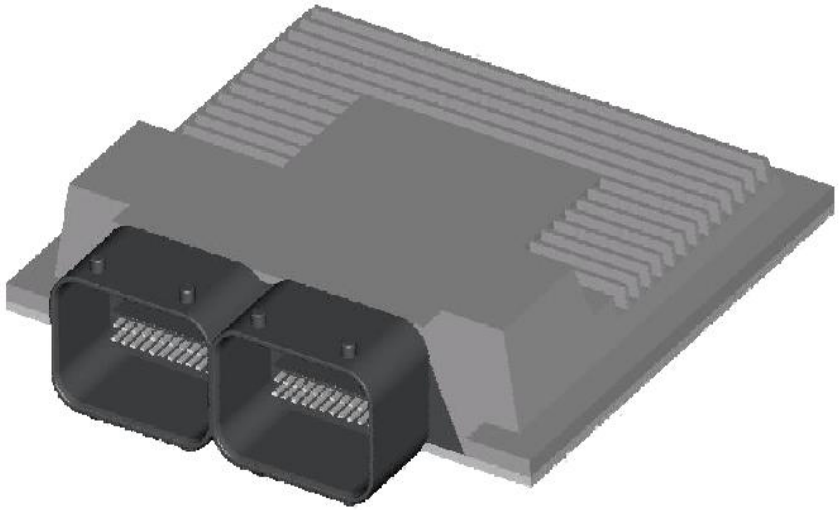


§4.5 ECU的优化设计 /Optimize Design of ECU

❑ ECU的可靠性和散热设计/ Reliability and Heat Dissipation Design

Mounting position	Passenger comp.	Engine comp.	On engine	In transmission
Temperature	-40 ... 85 °C	-40 / 105 °C	-40 / 105 ... 125 °C	-40 / 125 ... 150 °C
Vibration	... 5 g	5 ... 30 g	30 g ...	30 g ...





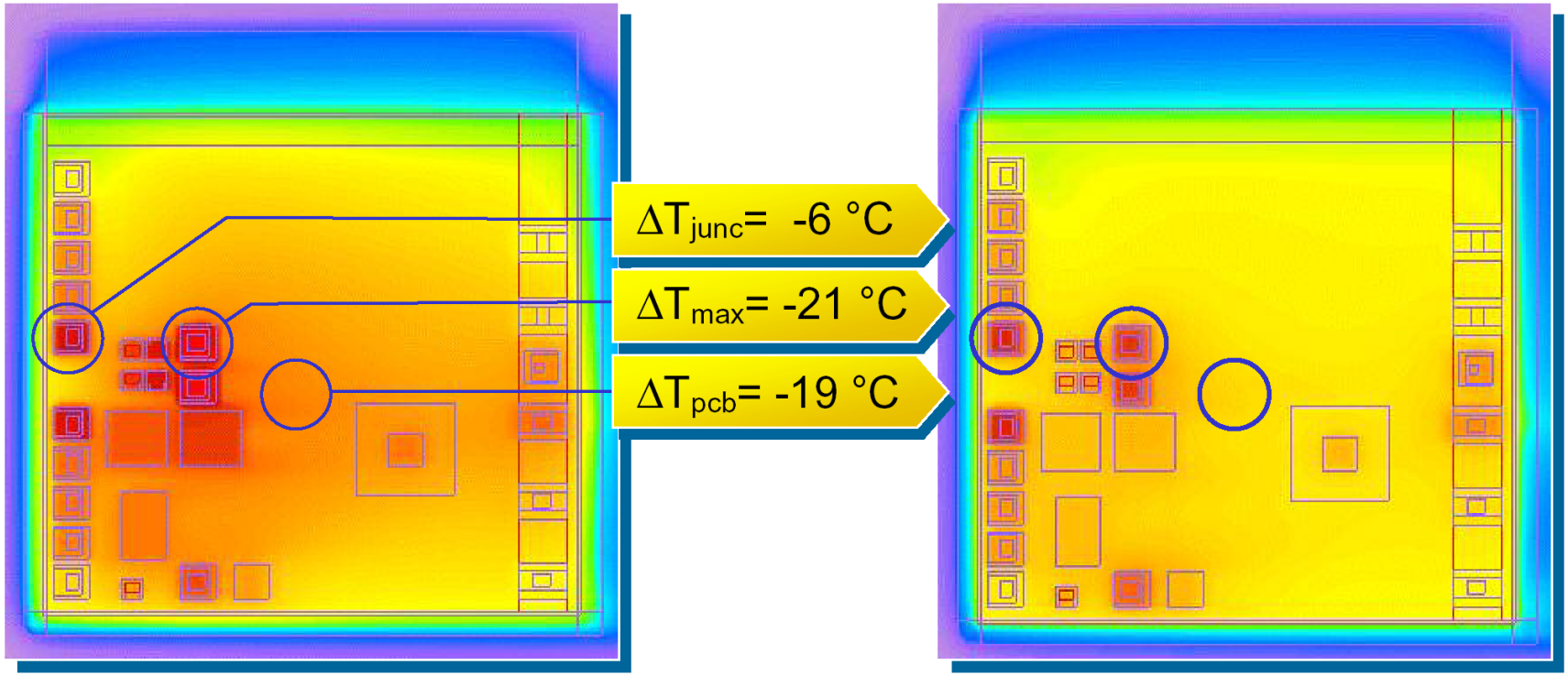


§4.5 ECU的优化设计 /Optimize Design of ECU

❑ ECU的可靠性和散热设计/ Reliability and Heat Dissipation Design

FR4 unsupported substrate

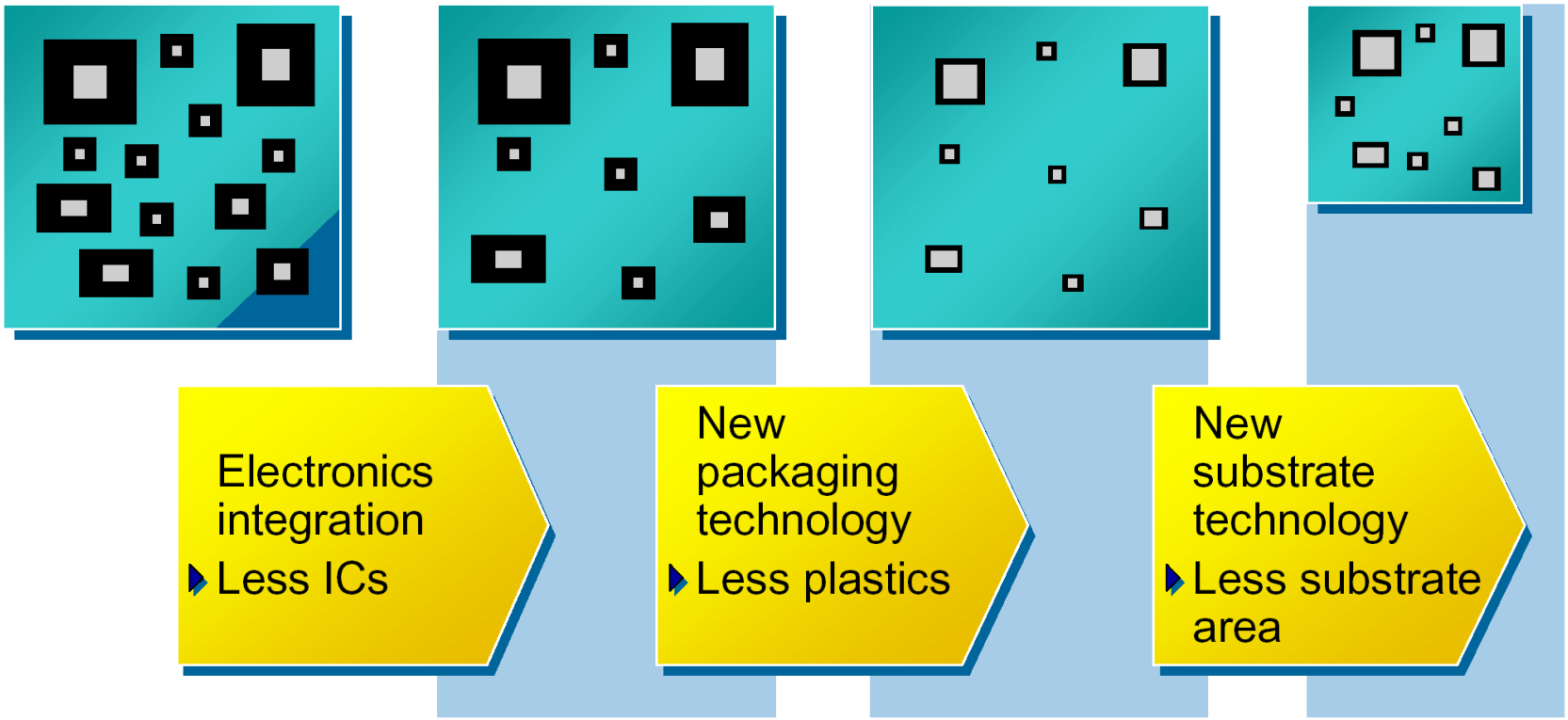
FR4 substrate laminated on aluminum





§4.5 ECU的优化设计 /Optimize Design of ECU

- ECU的体积优化：器件和工艺的改善/ Optimize the Volume of ECU: Improve the component and process





§4.5 ECU的优化设计 /Optimize Design of ECU

❑ECU的改进设计对比/ ECU Comparison

FR4

- 756 comp. (408 R's total)
- 6 layers (1 GND)

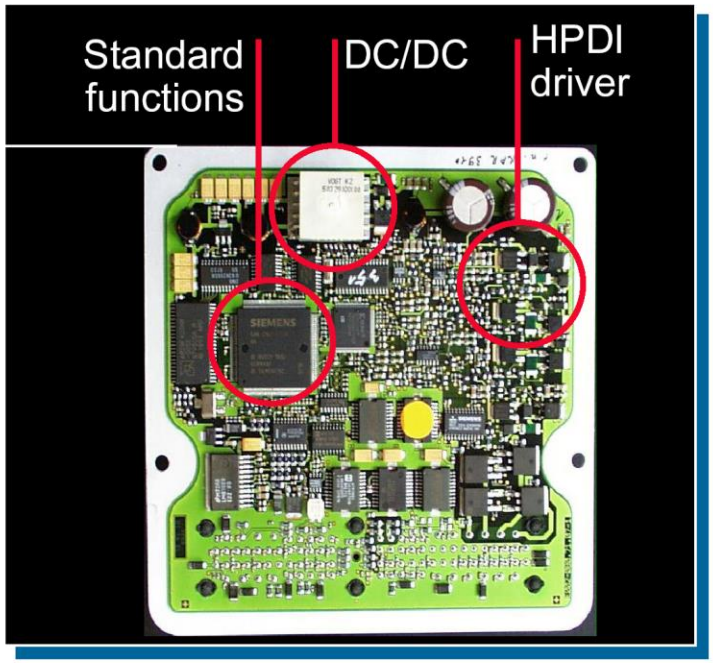


Size reduction

Up to 35 % reduction in layout area (depending on connector/PCB ratio)

SBU

- 756 comp. (408 R's total, 94 buried)
- 5 layers (1 GND, 1 layer buried R's)

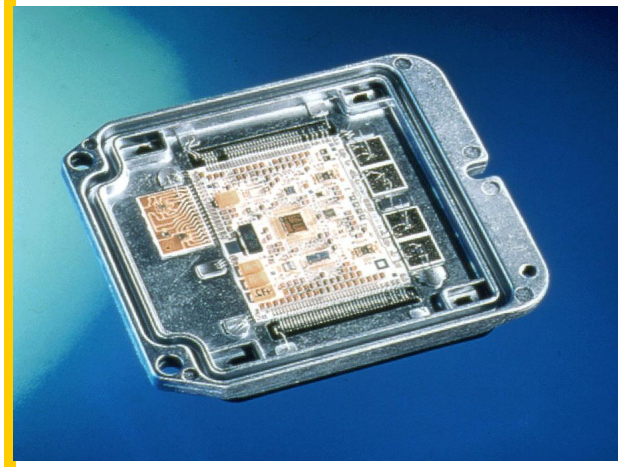




§4.5 ECU的优化设计 /Optimize Design of ECU

❑ 动力系统的ECU/ Photos of powertrain ECU

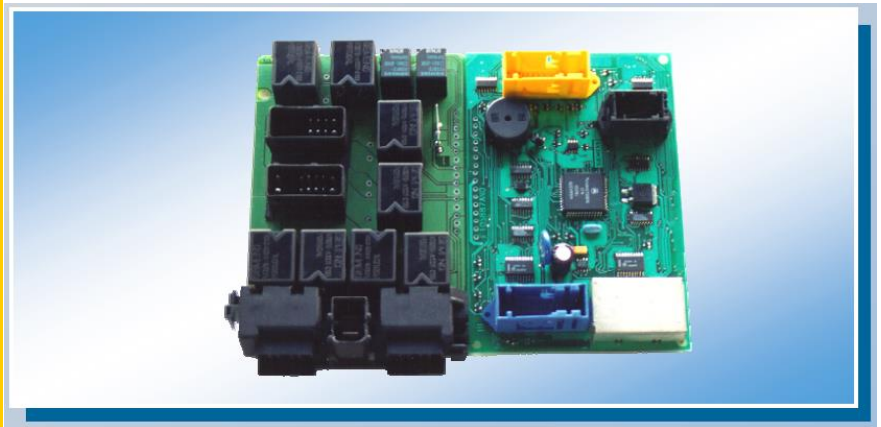


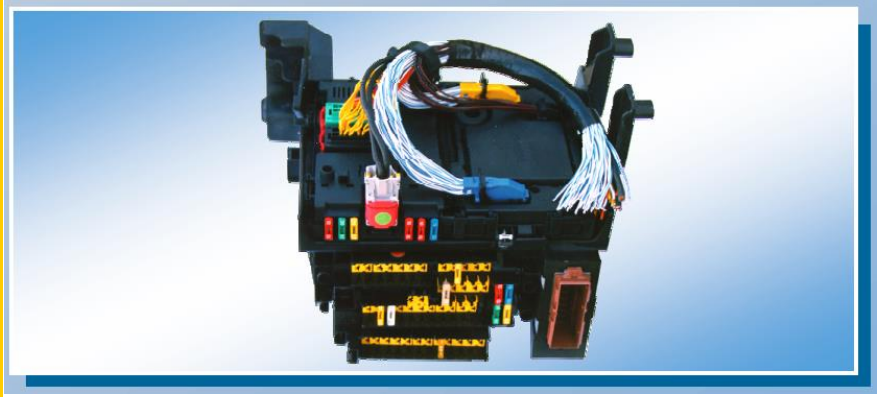


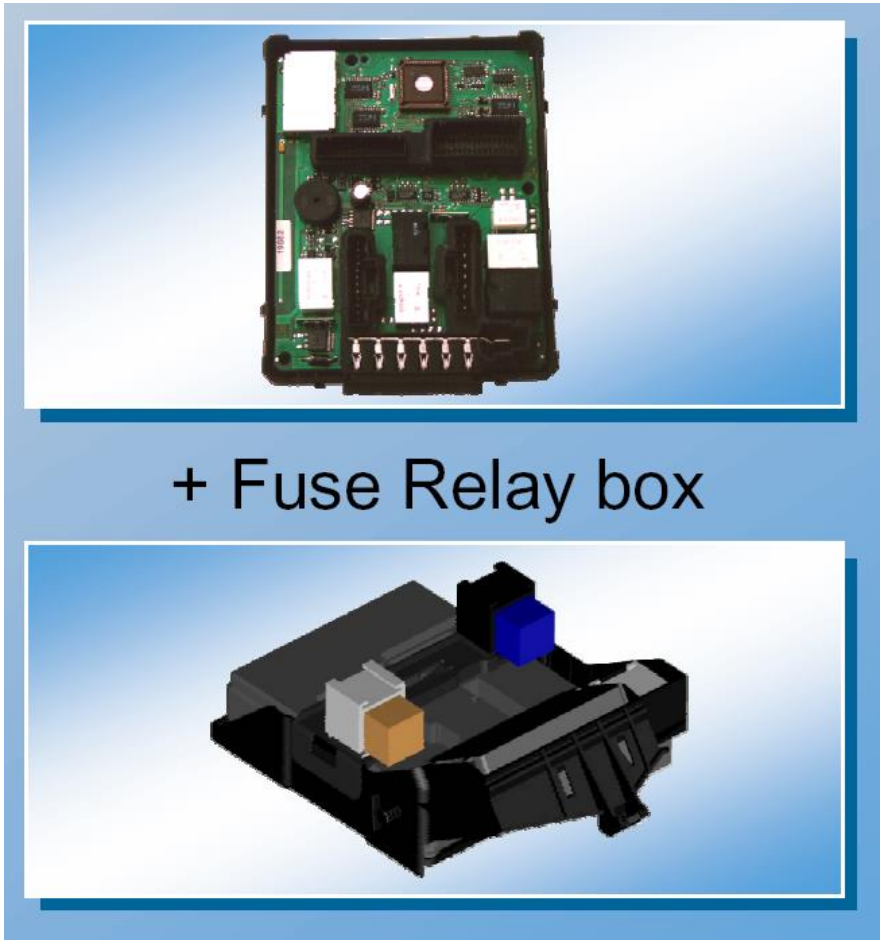


❑ 车身BCM/ Photos of body control module



+ Fuse Relay box

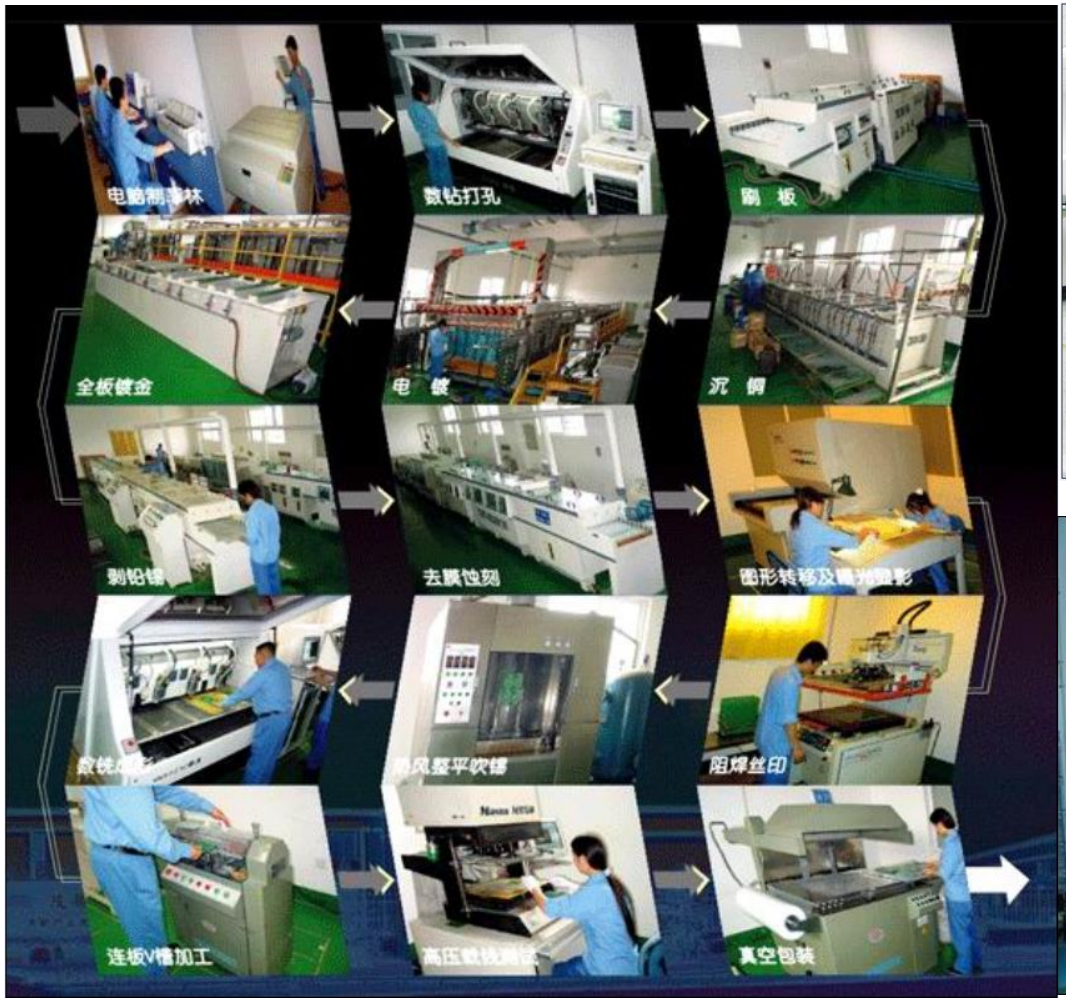






§4.6 ECU的生产和质量控制 /ECU Manufacture and Quality Management

1、电路板的制作



2、贴片机贴片



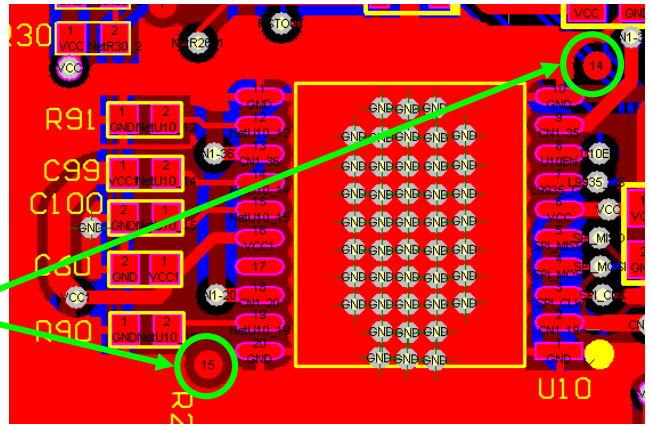


3、光学检测贴片位置是否正确

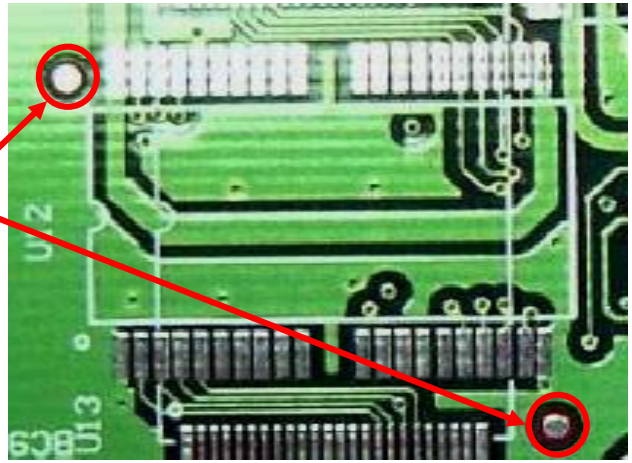


AOL自动光学检测仪

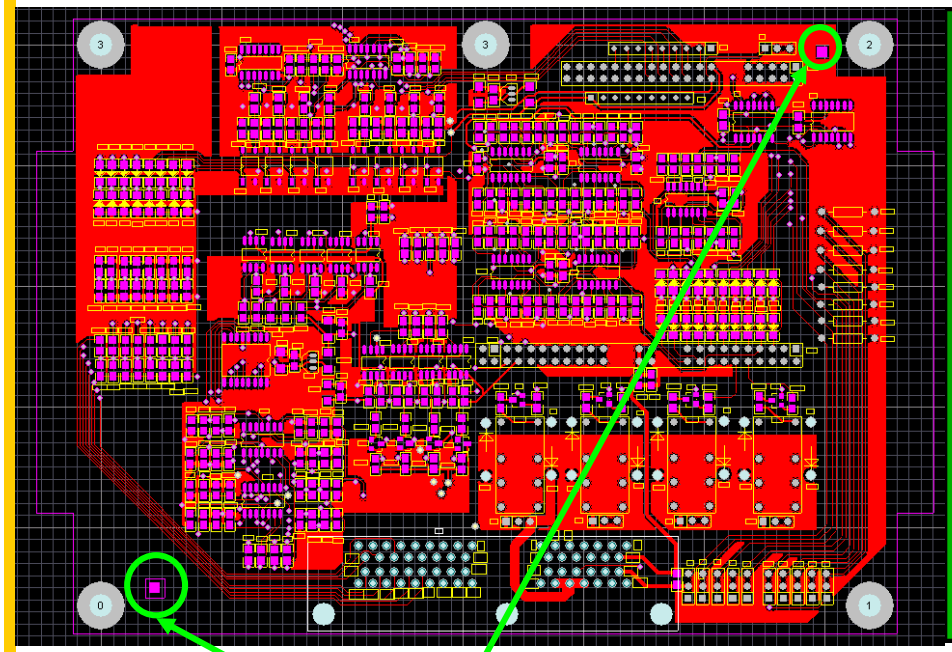
光学检测基准

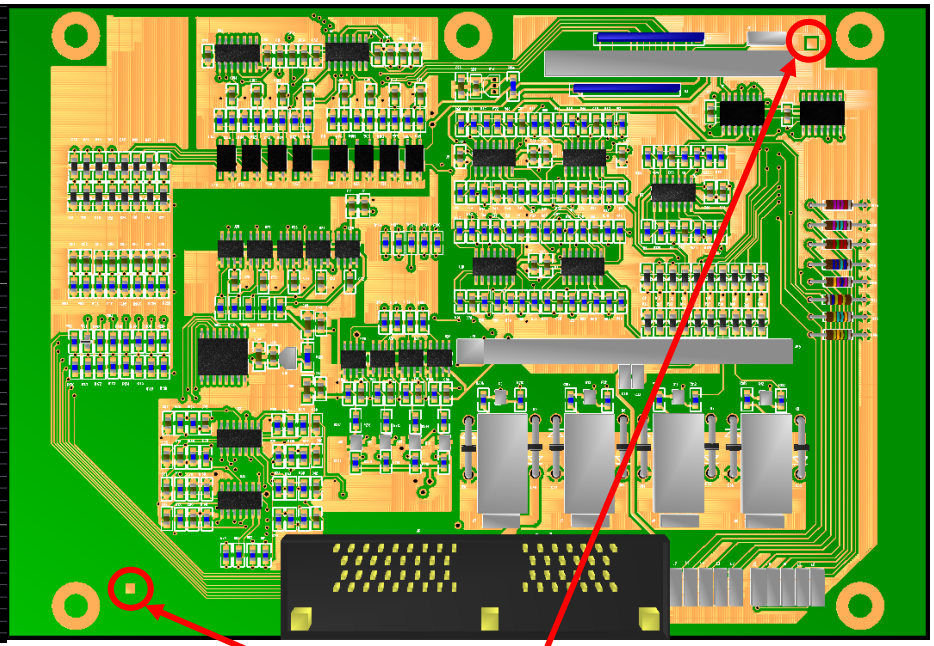


光学检测基准



3、光学检测贴片位置是否正确





光学检测基准

光学检测基准

4、回流焊



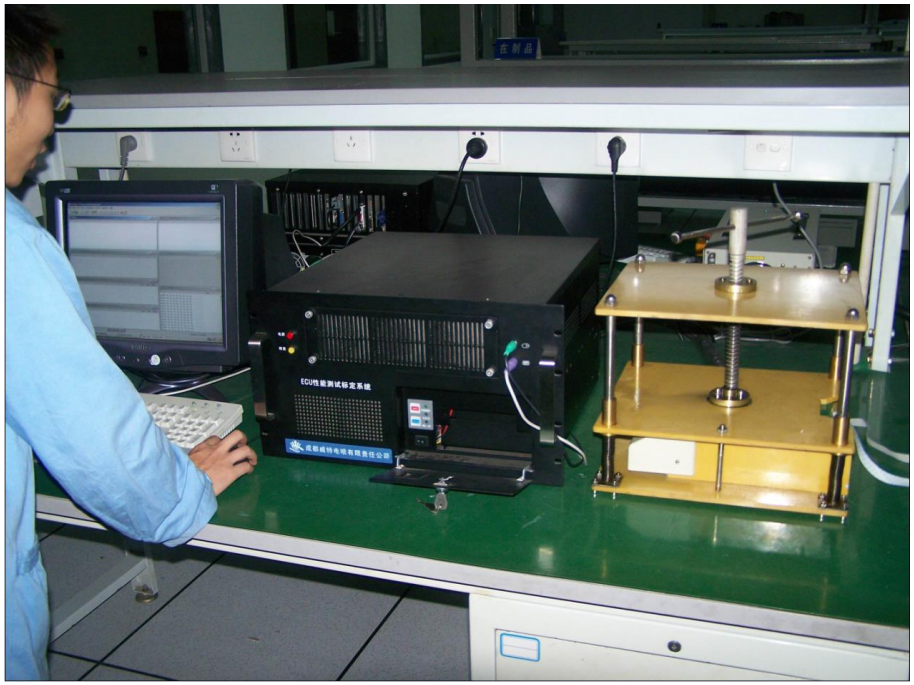


5、高温老化





6、针床测试和烧写程序

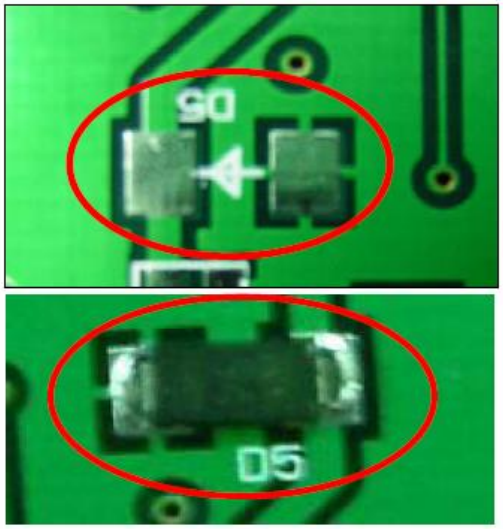
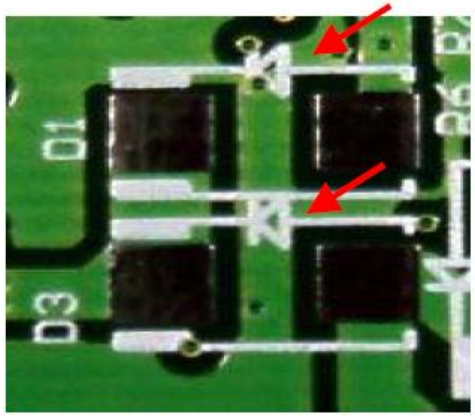




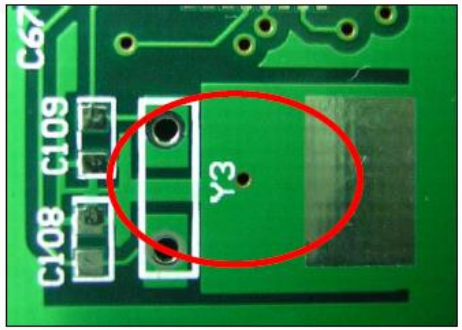
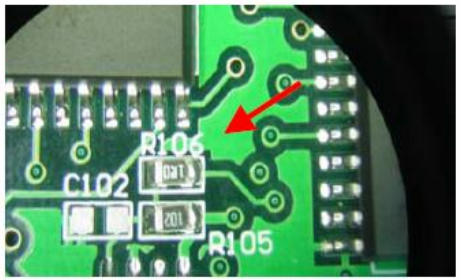
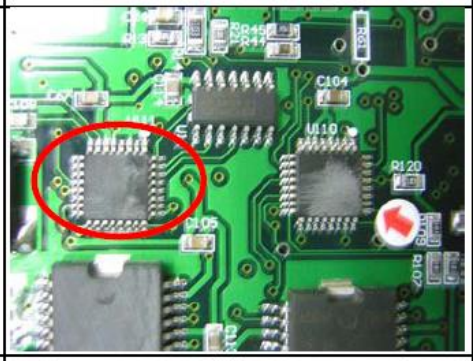
7、ECU功能检测和封装



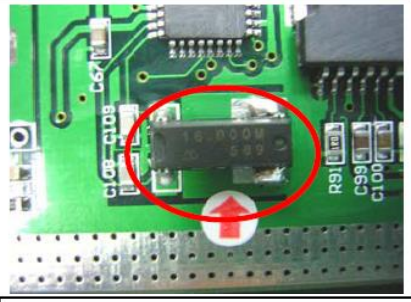
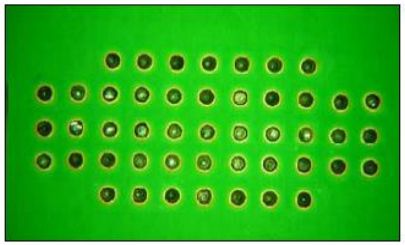
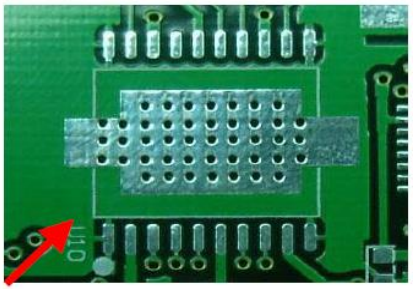
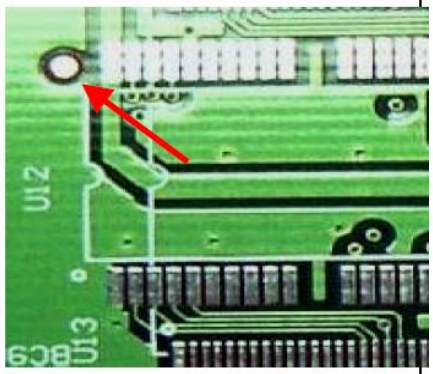
8、ECU加工过程的常见故障

項次	問題說明	不良現象	改善對策
1	材料尺寸與 PCB layout 尺寸不符，U112 元件腿跨距較大,與PCB Layout PAD不匹配		修正零件或設計
2	極性標示位置不恰當,不利於生產目檢作業. D1~D7 極性標識在元件PAD點中間，於上件後不便於目檢		 建議LAYOUT將極性標示放在元件側面

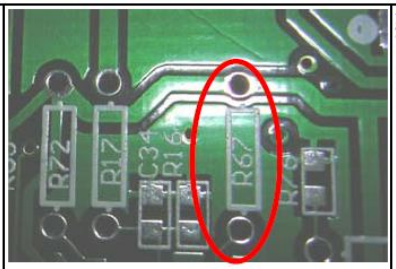
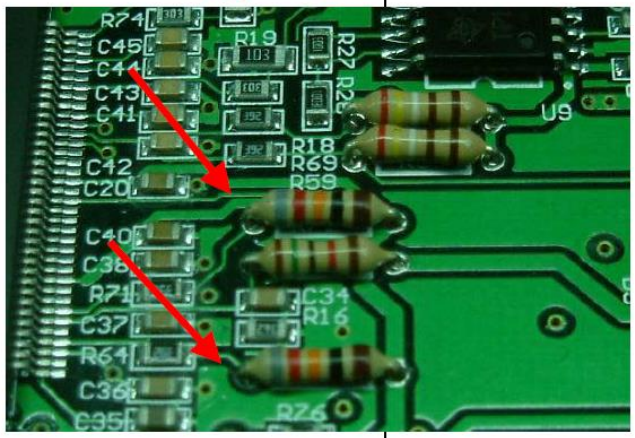
8、ECU加工过程的常见故障

3	PCB板上,元件位置標識在COMPONENT中間,不便於完成後目檢 Y3,Y1,OP1,C10,C11,C9,R58,R57,R70,R60,R72,R17,R67,R38,R53,R54,R61		建議LAYOUT將極性標示放在元件側面
4	SAMPLE的PCB板與現在生產產品版本不一樣, SAMPLE Board上有R106,現此次生產的PCB上無此元件位置.		建議 PCB 要有 Revsion 版本
5	元件無法辨識,易造成生產混料 U110&U111,U10&U8C元件外觀一樣,沒有標識進行辨別		建議在元件上做出標識 Mark

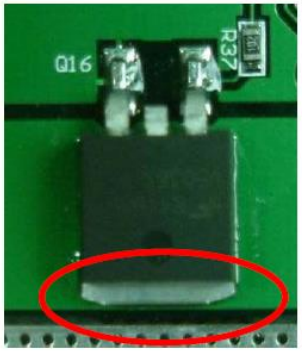
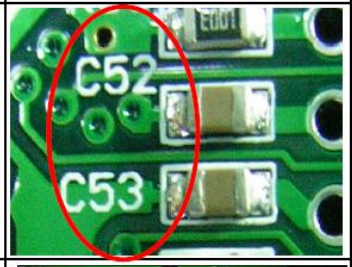
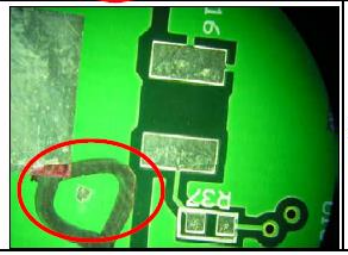
8、ECU加工过程的常见故障

6	Y3 爲SMD零件,但PCB Layout 爲DIP方式		建議將DIP孔變爲SMD點,或者將元件變爲插件元件
7	U8A,U8B,U8C,U10 導穿孔未堵防焊綠漆,導致過爐後有錫珠		Solder side 建議用防焊綠漆塞孔
8	PCB和IC上無光學定位點		建議 LAYOUT 時加上光學定位點,以提供 SMT 座標定位作業. 

8、ECU加工过程的常见故障

9	PCB板上,手插電阻的DIP孔孔距偏小,易造成插件本體破裂	 	建議變更設計

8、ECU加工过程的常见故障

10	Q16,Q17,Q18 PAD點偏小,導致無法爬錫		建議PAD點外擴
11	兩次來料不一致(第一次來料7.3N,第二次來料8.2N,最新BOM爲8.2N) C76,C70,C77,C78,C79,C72,C73,C74,C92,C95,C96,C97,C82,C83,C98,C85,C86,C87,C88,C89,C49,C50,C51,C53,C54, C57,C58,C59,C52,C8,C56兩次來料不一致		建議來料與BOM一致
12	PCB板上有漏銅現象		要求PCB供應商改善